BARRIERS TO GLOBAL CAPITAL ALLOCATION*

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Observed international investment positions and cross-country heterogeneity in rates of return to capital are hard to reconcile with frictionless capital markets. This article develops a theory of international capital allocation: a multi-country dynamic spatial general equilibrium model in which the entire network of cross-border investment is endogenously determined. Our model features cross-country heterogeneity in fundamental risk, a demand system for international assets, and frictions that cause segmentation in international capital markets. We measure frictions affecting international investment and apply our model to data from nearly 100 countries, using a new dataset of international capital taxes and cultural, linguistic, and geographic distances between countries (geopoliticaldistance.org). Our model performs well in reproducing the composition of international portfolios, the cross section of home bias and rates of return to capital, and other key features of international capital markets. Finally, we carry out counterfactual exercises: we show that barriers to international investment reduce world output by 7% and raise the cross-country dispersion of capital per employee, contributing in a meaningful way to global inequality. *JEL codes:* E22, E44, F2, F3, F4, G15, O4.

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I. Introduction

International investment positions have greatly increased in recent decades, with total global cross-border positions now exceeding twice the world's GDP, up from just one-fifth in 1980 (Florez-Orrego et al. 2024). Understanding how these international investment positions emerge and evolve over time is a central question of the open-economy literature in macroeconomics.

Early work on this topic sought to develop two-country models that could rationalize net international investment positions (Obstfeld and Rogoff 1995). Subsequent empirical work noted that countries' multilateral gross positions were much larger than net positions, and that there is significant heterogeneity in countries' risk exposures (Lane and Milesi-Ferretti 2001). This led to the development of a second generation of models with risk premia, where international assets were imperfectly substitutable. The latter models could rationalize gross positions (Gourinchas and Rey 2007; Pavlova and Rigobon 2007; Maggiori 2017; and others).

Recent empirical work, however, has uncovered rich variation in bilateral gross investment positions that can be traced back in part to fiscal and regulatory barriers (Coppola et al. 2021). It has also shown the existence of large unexplained differences in countries' rates of returns to capital (Monge-Naranjo, Sánchez, and Santaeulàlia-Llopis 2019) and even systematic failures of international arbitrage parity relations (Du and Schreger 2022). All of this evidence points strongly to international investment frictions as a major force shaping global capital allocation. Consequently, there is an ongoing research effort to build open-economy models with segmented capital markets (Gabaix and Maggiori 2015; Itskhoki and Mukhin 2021; Maggiori 2022).

Despite such progress, several aspects of international investment remain poorly understood. One crucial issue concerns the spatial allocation of capital (i.e., the direction of international bilateral investments) and its relationship to international investment frictions and the marginal product of capital in each country. In particular, a comprehensive workhorse spatial model of international capital allocation, comparable to the gravity models of international trade (Eaton and Kortum 2002), has yet to emerge.

The aim of this article is to fill this gap. We propose a new quantitative theory of global capital allocation: a multi-country

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dynamic spatial equilibrium model, in which the entire network of bilateral investment positions, as well as countries' output and returns to capital, are endogenously co-determined. We bring this model to the data, using a novel dataset of international capital taxes and geopolitical distances between countries.

Our model serves two key purposes: the first is to reproduce and reconcile several key empirical features of international capital markets (among which are bilateral positions, rates of return heterogeneity, and home bias). The second is to enable structural counterfactuals: our framework allows us to simulate how removing or adding barriers affects production and investment in each country and to quantify the economic importance of international investment frictions. To achieve this, our model embeds a demand system for international assets that depends not only on countries' risk and return fundamentals but also on investment frictions faced by international investors, which vary depending on the destination country. Our logit asset demand system admits various microfoundations, including one based on rational inattention and information frictions, and another based on extreme value theory and asset transaction costs.

Bilateral capital allocations are central to our analysis for three reasons. First, they are interesting in and of themselves as indicators of countries' mutual exposures, in terms of financial risk and geopolitical leverage (Clayton, Maggiori, and Schreger 2024). Second, they reveal important patterns of market segmentation that would be obscured in aggregate data. Indeed, in the absence of frictions, our theory—like the capital asset pricing model (CAPM)—predicts that investors from different countries would hold identical portfolios. Yet we observe a markedly different pattern once frictions are introduced. Third, bilateral variation is crucial to our empirical strategy: it allows us to identify key model parameters by estimating a theory-grounded gravity regression of international asset positions, controlling for all origin and destination fixed effects.

Our framework predicts, as a consequence of international investment barriers, the emergence of a core-periphery structure in international capital markets. Portfolios are skewed toward countries that are easily accessible to international investors: these "central" countries display high capital stocks and low rates of return to capital. By contrast, "peripheral countries" that are less accessible to international investor have lower capital stocks and higher rates of return, compensating investors for

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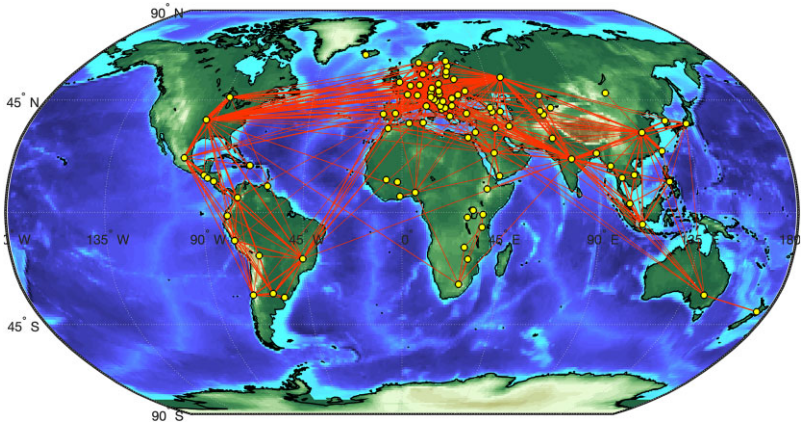

FIGURE I

Equilibrium Spatial Investment Network

The figure plots the model-implied equilibrium cross-border investment as a network. Each node is a country, and the link thickness reflects the geometric average of the (i, j) and (j, i) investment position ($\sqrt{a_{ij}a_{ji}}$). Minor links ($<$ \$10 bln) are omitted.

overcoming those barriers. That is, rates of return reflect not only risk premia but also countries' accessibility to international investors. The resulting global allocation of capital is generally suboptimal. Figure I illustrates the model-implied equilibrium cross-border investment as a network.

In our empirical analysis, we focus on three types of frictions: (i) capital income taxes, (ii) political risk, and (iii) geographic, linguistic, and cultural distances between countries, whose effect on international portfolios can be estimated by running a gravity regression on bilateral investment data.[1]

In measuring these frictions, we make a data contribution: we develop a new suite of bilateral indicators of international capital taxation and geopolitical distances across countries, made available to other researchers at www.geopoliticaldistance.org. This repository includes new measures of cultural, linguistic, and geographic distance between countries, which greatly improve on

1. The main difference with previous empirical gravity studies is that our gravity regression has a structural interpretation: it is derived from the model, and we can use it to recover model parameters—namely, the elasticities of the portfolio shares with respect to distances.

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the previous generation of indicators in terms of coverage and measurement (Spolaore and Wacziarg 2016).

Taking our model to the data, we obtain four sets of empirical results. First, we find that cultural, linguistic, and geographic distances generate a strong gravity effect on international portfolios. The effect is quantitatively similar for different subcategories of foreign investment: equity versus debt, foreign direct investment versus foreign portfolio investment. Our measured elasticities condition on origin-country and destination-country fixed effects, are robust to including a large set of control variables, and remain quantitatively large irrespective of the estimation method.

Second, exploiting the breakdown between residency-based and nationality-based bilateral investment data (Damgaard, Elkjaer, and Johannesen 2019; Coppola et al. 2021), as well as the difference between de jure and de facto tax rates (made possible by our new bilateral tax dataset), we show that the share of securities issued through tax havens is robustly predicted by the reduction in withholding tax rates offered by tax havens. This supports the view that tax incentives are an important driver of the discrepancy between residency-based and nationality-based bilateral investment positions.

Third, we find that our model, taken to the data, produces realistic country portfolios and rates of return to capital, and predicts with striking accuracy (out of sample) the home bias of individual countries.[2] The model also predicts that emerging economies and countries with a higher degree of home bias should exhibit higher rates of return on capital. These predictions are consistent with existing empirical evidence (Lau, Ng, and Zhang 2010; David, Henriksen, and Simonovska 2014). Another notable finding is that our model can predict (out of sample) some of the differences between the nationality-restated and the residency-based investment data, allowing us to cross-validate our methodological approach against that of Coppola et al. (2021) and Damgaard, Elkjaer, and Johannesen (2019).

Fourth, we carry out a counterfactual analysis, using the model to study the quantitative implications of removing barriers to global capital allocation. We find that our estimated barriers introduce significant capital misallocation across countries.

2. We say "out of sample" because we estimate our gravity equation without using any data on domestic investment.

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Compared to a situation without barriers to global capital allocation, world GDP is about 7% lower. An important result is that barriers to capital movements contribute significantly to cross-country inequality. We find that the standard deviation of log capital per employee is 50.9% higher than it would be in a world without barriers, and the dispersion in output per employee is 22.0% higher.

The largest gains from removing barriers would accrue to developing countries, predominantly in Africa, Asia, and Latin America: the investment barriers in our model make these countries less accessible to international investors. In a counterfactual scenario with no barriers, these countries would become significant net receivers of capital. By contrast, richer countries would become net providers of capital. This is very different from the baseline (distorted) model, where net international investment positions are mostly uncorrelated with income levels. Thus, our analysis can account for the dearth of investment from rich to poor countries observed by Lucas (1990).

We extend the basic model along several dimensions, by incorporating various additional frictions: barriers to goods trade, capital controls, and currency hedging costs (to capture currency risk). None of these extensions materially affects our headline findings. We also perform a wide range of robustness checks.

To summarize our contribution: we provide and empirically estimate a multi-country spatial model of bilateral international investment with investment barriers, using new data on bilateral tax rates and geopolitical distances between countries. Our analysis reconciles several stylized facts about the global allocation of capital and allows us to study how the global economy reacts to changes in international investment frictions. We find that these barriers have important effects for the distribution of capital across countries, efficiency, and global inequality.

This article contributes to the long-standing literature in open-economy macroeconomics that seeks to understand international investment positions, starting with Obstfeld and Rogoff (1995), continuing with Pavlova and Rigobon (2007), Gourinchas and Rey (2007), Maggiori (2017), and others, and more recently with models of segmented capital markets (Gabaix and Maggiori 2015; Itskhoki and Mukhin 2021; Maggiori 2022).

In addition, this study is made possible by publicly available data on international investment positions that have been assembled and dissected over the years by many researchers (Lane and

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Milesi-Ferretti 2001; Damgaard, Elkjaer, and Johannesen 2019; Maggiori, Neiman, and Schreger 2020; Coppola et al. 2021; Beck et al. 2024).

Although our approach is structural, it bears a clear connection to an earlier empirical literature that used gravity-type regressions to study international financial variables. Some salient examples of this approach include Di Giovanni (2005), Lane and Milesi-Ferretti (2008), and Portes and Rey (2005). Notably, the latter provided an interpretation of gravity patterns in terms of information frictions.

Theoretical models that sought to capture some of these gravity effects include those of Martin and Rey (2004) and more recently Gârleanu, Panageas, and Yu (2020). An early contribution that combined theory and data is Head and Ries (2008), who focused specifically on cross-border mergers and acquisitions. Last, recent work documented gravity effects in exchange rates return factors (Lustig and Richmond 2017, 2020).

We also build on previous work on natural resources and capital misallocation by Monge-Naranjo, Sánchez, and Santaeulàlia-Llopis (2019), David, Henriksen, and Simonovska (2014), and earlier work by Caselli and Feyrer (2007). We incorporate natural resources explicitly in our theory and dataset, ensuring that our model-based estimates of marginal product of capital is consistent with this previous work, while using the most up-to-date available data (Feenstra, Inklaar, and Timmer 2015, Penn World Table 10, World Bank 2018). Consistent with the more recent findings by Monge-Naranjo, Sánchez, and Santaeulàlia-Llopis (2019) and David, Henriksen, and Simonovska (2014), which differ from the original estimates by Caselli and Feyrer (2007), our model generates large and persistent differentials in capital returns across countries, implying that capital is not efficiently allocated across countries.[3]

A related line of research (Barro, Mankiw, and Sala-i-Martín 1995; Colacito and Croce 2010) studies how capital mobility can speed up the process of convergence to the steady state in a neoclassical framework and quantifies the resulting welfare gains. Gourinchas and Jeanne (2006, GJ) found these welfare gains to

3. Our analysis also differs from Ohanian, Restrepo-Echavarria, and Wright (2018), who focused on international capital misallocation stemming from labor market distortions that may lower investment incentives by reducing the supply of labor.

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be small. Our findings of large income and welfare effects from global capital misallocation do not contradict GJ: while they focus on the speed of adjustment along the transition path, we focus on mechanisms that generate persistent capital misallocation, even in the steady state.

In terms of modeling techniques, this article relates to the recent literatures on dynamic spatial equilibrium models (Eaton et al. 2016; Ravikumar, Santacreu, and Sposi 2019; Liu and Ma 2021; Kleinman, Liu, and Redding 2023; Liu and Tsyvinski 2024); asset demand systems (Koijen and Yogo 2019, 2020); and rational inattention and information frictions, particularly in international finance (Van Nieuwerburgh and Veldkamp 2009; Dziuda and Mondria 2012; Matějka and McKay 2015). Because one of the microfoundations we provide for our asset demand system is based on rational inattention, our main technical innovation with respect to this earlier literature on information frictions and international finance is that we are the first to embed information frictions into a quantifiable multi-country model.

We connect to previous work on the institutional, historical, and cultural determinants of international investment positions (Papaioannou 2009; Aggarwal, Kearney, and Lucey 2012; Ahern, Daminelli, and Fracassi 2015; Burchardi, Chaney, and Hassan 2019). More broadly, our study relates to the literature on historical and cultural barriers to international exchanges (Guiso, Sapienza, and Zingales 2009; Felbermayr and Toubal 2010) and the spread of innovations and development across countries (Spolaore and Wacziarg 2009, 2012, 2016, 2018).

II. A Spatial Model of International Capital Allocation

II.A. Firms

In this section we present our dynamic spatial model of international capital allocation. We start from the production side. Time is discrete and indexed by t. There are n countries indexed by $i \in \{1, 2, , \ldots, n\}$. Each country has a representative firm that produces a homogeneous, tradable good using a three-factor Cobb-Douglas production function.

Total output is stochastic and equal to

$$y_{it} = \zeta_{it} \cdot \omega_i \cdot k_{it}^{\kappa_i} \cdot \ell_{it}^{\lambda_i} \cdot x_{it}^{\xi_i}, \tag{1}$$

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where ω_i is country i's (expected) total factor productivity; k_{it} is the input of reproducible capital; ℓ_{it} is the labor input; x_{it} is the input of natural resources.[4] The parameters κ_i, λ_i, and ξ_i, which are equal to the equilibrium income shares of reproducible capital, labor, and natural resources (respectively), are allowed to vary across countries. The production function satisfies constant returns to scale ($\kappa_i + \lambda_i + \xi_i = 1$). ζ_{it} is a log-normally distributed shock to output with expectation equal to one.

We construct the shock ζ_{it} as follows: its log can be decomposed as the sum of a country-level shock z^c_{it} and a global (country-invariant) shock z^w_t. Both follow a Gaussian distribution:

$$\log \zeta_{it} \stackrel{\text{def}}{=} z^c_{it} + z^w_t, \qquad z^c_{it}, z^w_t \sim N. \tag{2}$$

We denote by σ^2_i the variance of z^c_{it}, and by σ_{iw} the covariance of z^c_{it} and z^w_t. Therefore, each country's output is allowed to covary with the global economic cycle to a different degree.

Labor and natural resources cannot be moved across countries. Capital is the only mobile factor. Each unit of the final good can be used for consumption or saved and transformed into capital to be used for production in the next period. Capital used for production incurs a random depreciation rate $(1 - \delta_{it})$ that is equal (by assumption) to an expected value equal to $(1 - \delta)$ times the total factor productivity (TFP) shock ζ_{it}. Hence the value of repatriated capital is also affected by the business cycle.[5]

The global resource constraint is thus:

$$\sum_{i=1}^{n} \left[c_{it} + k_{it+1}\right] = \sum_{i=1}^{n} \left[y_{it} + (1 - \delta_{it})\, k_{it}\right], \tag{3}$$

where c_{it} is current period consumption of country i's agents (including the government). The final homogeneous good is assumed to be the numéraire of the economy (its price is normalized to one).

The representative firm issues shares, which entitle capital investors to a proportion of the firm's residual capital income. The

4. Natural resources are included in our theoretical and empirical analyses in line with contributions by Caselli and Feyrer (2007) and Monge-Naranjo, Sánchez, and Santaeulàlia-Llopis (2019).

5. This is a simplified way of modeling volatility in the value of financial assets.

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firm maximizes capital income:

$$\max_{\ell_{it}, x_{it}} y_{it} - w_{it}\ell_{it} - m_{it}x_{it}. \tag{4}$$

The equilibrium rental rate of natural resources (m_{it}) and wage rate (w_{it}) are determined in competitive markets as usual:

$$m_{it} = \xi_i \frac{y_{it}}{x_{it}}; \qquad w_{it} = \lambda_i \frac{y_{it}}{\ell_{it}}. \tag{5}$$

The capital income share is therefore equal to κ_i. Because capital is a mobile factor, different shares of this income will accrue to investors from different countries. In what follows, we use index i to refer to the country where production takes place (the destination country), and index j to refer to the country that provides the capital (the investor country).

Each destination country i exogenously imposes a tax rate equal to a share $(1 - \tau_{ij})$ of the capital income accruing to agents in origin country j. We clarify that τ_{ij} does not capture expropriation risk; we model and quantify that using a separate wedge.

A unit of capital invested in country i yields a pretax income that is equal to profits per unit of capital or (equivalently) the marginal product of capital in country i, which is defined as:

$$\mathrm{MPK}_{it} \stackrel{\text{def}}{=} \kappa_i \frac{y_{it}}{k_{it}}. \tag{6}$$

In addition, investors receive back their proportional share of undepreciated capital at the end of the investment period. We call r_{ijt} the (gross, after-tax) return on a unit of capital invested in country i from country j. It is by definition equal to:

$$r_{ijt} = \zeta_{it}\left[1 + \tau_{ij}\,\mathbb{E}_{t-1}(\mathrm{MPK}_{it}) - \delta\right]. \tag{7}$$

II.B. Households

Each country is populated by a dynasty of infinitely lived households, who are born as workers, work for one period, transition to being capitalists in period two, and from then on every period may die with some random probability $\mathfrak{D}$. Bequests are redistributed to the newly born cohort.[6]

We denote by (j, h) the cohort that is born at time h in country j. Each cohort has unit mass at birth, and its mass gradually shrinks exponentially by $\mathfrak{D}$. At birth ($t = h$), each cohort (j, h) is

6. Whether the agents have a bequest motive will be incorporated in their discount rate, but in any case it is immaterial for our analysis.

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endowed with ℓ_j units of labor; it also earns income from government payments, which include the rents from natural resources $m_{jt}x_j$ and a government transfer T_{jt} which includes rebated tax revenues as well as inherited wealth from the agents from the previous cohorts that died in the last period.

We denote the consumption per capita of cohort h of country j at time t as C_{jht} and we denote their saved wealth S_{jht}. We denote their maximized recursive utility as $V_{jht}(S_{jht-1})$:

$$(8)\quad V_{jht}\left(S_{jht-1}\right) \stackrel{\text{def}}{=} \max_{\{C_{jh}\}^t\{S_{jh}\}^t} \log C_{jht} + \theta_j\, \mathbb{E}_{jht}\left[V_{jht+1}\left(S_{jht}\right)\right],$$

where θ_j is a country-specific patience parameter (which already incorporates the probability of death). The maximization is subject to the following budget constraint:

$$(9)\qquad C_{jht} + S_{jht} = \begin{cases} w_{jt}\ell_j + m_{jt}x_j + T_{jt} & \text{if} \quad t = h \quad \text{(Workers)} \\ R_{jt}\, S_{jht-1} & \text{if} \quad t > h \quad \text{(Capitalists)}, \end{cases}$$

where R_{jt} is the gross return on the portfolio of securities that j-households hold from time $t-1$ to t. For expositional purposes, we assume that the capitalists delegate their portfolio formation to an international financial intermediary and thus take R_{jt} as given (this assumption can be relaxed, see Section II.D). For $t > h$, the right-hand side of the budget constraint is thus the capitalists' wealth that has been accumulated at time t.

We begin by solving for the capitalists' consumption and saving decision. Their Euler equation takes the usual form:

$$(10)\qquad \theta_j\, \mathbb{E}_t\left(\frac{C_{jht}}{C_{jht+1}} \cdot R_{jt+1}\right) = 1.$$

After substituting saved wealth into the Euler equation, we can guess and verify that, in equilibrium, all investors save a constant share θ_j of their income (if workers) or wealth (if capitalists).

II.C. International Capital Markets

Let a_{ijt} be the total claims to country i capital by the investors of country j. Capital markets clearing implies that (i) the supply of physical capital to country i (k_{it}) equals the sum of all units of capital supplied from all countries j; (ii) total claims by country j towards all countries i must equal country j's total savings from the previous period:

$$(11)\qquad k_{it} = \sum_{j=1}^{n} a_{ijt}; \qquad s_{jt-1} = \sum_{i=1}^{n} a_{ijt}.$$

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Let us define π_{ijt} as the share of $j-$capital invested in country i as a percentage of country j's aggregate saving (the portfolio share), formally:

$$\pi_{ijt} \stackrel{\text{def}}{=} \frac{a_{ijt}}{s_{jt-1}}. \tag{12}$$

In matrix form, equation (13) describes market clearing for international capital (where capital is sourced and where it is used):

$$\underbrace{\begin{bmatrix} k_{1t} \\ k_{2t} \\ \vdots \\ k_{nt} \end{bmatrix}}_{\mathbf{k}_t} = \underbrace{\begin{bmatrix} \pi_{11,t} & \pi_{12,t} & \cdots & \pi_{1n,t} \\ \pi_{21,t} & \pi_{22,t} & \cdots & \pi_{2n,t} \\ \vdots & \vdots & \ddots & \vdots \\ \pi_{n1,t} & \pi_{n2,t} & \cdots & \pi_{nn,t} \end{bmatrix}}_{\mathbf{\Pi}_t} \underbrace{\begin{bmatrix} s_{1,t-1} \\ s_{2,t-1} \\ \vdots \\ s_{n,t-1} \end{bmatrix}}_{\mathbf{s}_{t-1}}. \tag{13}$$

II.D. Asset Allocation

Next we specify how international portfolios are formed. We assume that international investment is carried out by financial intermediaries that are perfectly competitive and who allocate investments using a logit asset demand system (Koijen and Yogo 2020). For expositional purposes, we assume that the asset demand system takes a specific functional form. In Online Appendix A, we provide multiple microfoundations for this asset demand function: we show that, under certain assumptions, households and investment intermediaries can be combined into a single agent, who allocates investment as follows.[7]

The share of j capital invested in destination country i by the intermediary is equal to:

$$\pi_{ijt} = \frac{\left(\frac{\mathcal{R}^{\eta}_{ijt-1} k_{it}}{\Delta_{ij}}\right)}{\sum_{\iota=1}^{n} \left(\frac{\mathcal{R}^{\eta}_{\iota jt-1} k_{\iota t}}{\Delta_{\iota j}}\right)} \tag{14}$$

where $\mathcal{R}_{ijt}$ is the risk-adjusted expected return to capital in country i, defined (in accordance with the microfoundations) as:

$$\log \mathcal{R}_{ijt} \stackrel{\text{def}}{=} \mathbb{E}_t(\log r_{ijt+1}), \tag{15}$$

7. In other words, the asset demand system below does not require any agency frictions between investors and the intermediary.

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and Δ_{ij} is what we call a "portfolio wedge," a distortionary term that captures various frictions affecting the investors' ability to invest from country j to country i. These portfolio wedges Δ_{ij} can be interpreted in multiple ways, depending on the chosen microfoundation (discussed in the next subsection).

By construction, $\mathcal{R}_{ijt}$ is equal to:

$$\mathcal{R}_{ijt} = \frac{1 + \tau_{ij}\, \mathbb{E}_t\left(\mathrm{MPK}_{it+1}\right) - \delta}{\exp\left(\frac{1}{2}\sigma_i^2 + \sigma_{iw}\right)} \tag{16}$$

and thus incorporates the physical productivity of capital (MPK), taxation (τ_{ijt}), and a risk premium (the denominator), which in turn depends on country i's productivity shock's variance and covariance with the global TFP shock.

II.E. Microfoundations for Asset Demand and Interpretation of the Portfolio Wedges Δ_{ij}

As mentioned, the asset demand system in equation (14) can be derived from different microfoundations, each offering a different interpretation of the portfolio wedges Δ_{ij}. These theories are discussed in detail in Online Appendix A. Here we provide a brief overview of these alternative microfoundations.

The first microfoundation is based on the rational inattention logit model of Matějka and McKay (2015) and the closed-form results of Pellegrino (2023). In this framework, investors face uncertainty about the risk-adjusted returns of the various destination countries and must learn about them: they must allocate optimally their limited attention across potential investment destinations. Under this microfoundation, the wedge Δ_{ij} reflects the precision of the prior of country j's investors' beliefs about the risk-adjusted rate of return in country i.

The second microfoundation rests on extreme value theory, which is widely used in the international trade literature.[8] This approach assumes that investors are subject to heterogeneous asset trade costs, with the noise following an extreme value distribution. In this context, Δ_{ij} can be interpreted as

8. In the April 2021 version of this article (Pellegrino, Spolaore, and Wacziarg 2021), we used extreme value theory as the microfoundation for our model of international portfolios. A more recent application of extreme value theory as a microfoundation for international investment positions can be found in Kleinman, Liu, and Redding (2023).

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capturing systematic transaction costs incurred to invest across borders.

Thirdly, in Online Appendix A we discuss under what additional assumptions we can invoke the microfoundations of Koijen and Yogo (2020), based on asset characteristics, in our setting.

These three microfoundations offer slightly different economic interpretations of the wedges Δ_{ij}, but they all lead to the same functional form for our asset demand system. In Section VII, we conduct counterfactual analysis on these portfolio wedges. The specific interpretation of Δ_{ij} affects how we view our counterfactual analysis. If Δ_{ij} represents information frictions, removing these wedges simulates a change in the investor's prior information. If they represent intrinsic trading costs, our counterfactual can be understood as simulating the effects of a hypothetical improvement in the asset trading technology.

Importantly, our counterfactual analysis remains informative about the potential gains from reducing barriers to international capital allocation, regardless of whether these barriers are viewed as inefficiencies or intrinsic costs. This approach mimics that of the recent literature on "universal gravity" in international trade (Allen, Arkolakis, and Takahashi 2020), which aims to produce quantitative estimates of the gains from trade and the effects of trade costs that are valid for a whole class of gravity models based on different microfoundations.

II.F. Global Capital Markets Clearing, Balance of Payments, and Steady-State Equilibrium

To close the model, we write country j's savings in terms of income:

$$s_{jt} = \theta_j \left[(\lambda_j + \xi_j) y_{jt} + \sum_{i=1}^{n} r_{ijt} \pi_{ijt} s_{jt-1} + \sum_{i=1}^{n} (1 - \tau_{ij}) \mathrm{MPK}_{jt} \pi_{jit} s_{it-1} \right]. \tag{17}$$

By defining the matrices

$$\mathbf{R} \stackrel{\text{def}}{=} \begin{bmatrix} r_{11} & r_{12} & \dots & r_{1n} \\ r_{21} & r_{22} & \dots & r_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ r_{n1} & r_{n2} & \dots & r_{nn} \end{bmatrix}; \quad \mathbf{\Theta} \stackrel{\text{def}}{=} \begin{bmatrix} \theta_1, & 0, & \dots & 0 \\ 0 & \theta_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \theta_n \end{bmatrix} \tag{18}$$

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$$\mathbf{\Lambda} \stackrel{\text{def}}{=} \begin{bmatrix} \lambda_1, & 0, & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}; \quad \mathbf{\Xi} \stackrel{\text{def}}{=} \begin{bmatrix} \xi_1, & 0, & \dots & 0 \\ 0 & \xi_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \xi_n \end{bmatrix}, \tag{19}$$

we can rewrite equation (17) in linear algebra notation as:

$$\begin{aligned} \mathbf{s}_t = \mathbf{\Theta} \big\{ &(\mathbf{\Lambda} + \mathbf{\Xi})\mathbf{y}_t + \left[(\mathbf{\Pi}_t \circ \mathbf{R}_t)'\,\mathbf{1}\right] \circ \mathbf{s}_{t-1} \\ &+ \left[\mathbf{\Pi}_t \circ \left(\mathbf{1}\mathbf{1}' - \mathbf{T}\right) \circ \mathbf{mpk}_t\,(\mathbf{k}_t)\,\mathbf{1}'\right] \mathbf{s}_{t-1} \big\}, \end{aligned} \tag{20}$$

where $\circ$ is the Hadamard (entry-by-entry) product. Notice that this equation can be rewritten as:

$$\begin{aligned} &\mathbf{c}_t + \mathbf{s}_t = \mathbf{y}_t \\ &+ \underbrace{\left[(\mathbf{\Pi}_t \circ \mathbf{R}_t)'\,\mathbf{1}\right] \circ \mathbf{s}_{t-1} - \boldsymbol{\kappa} \circ \mathbf{y}_t + \left[\mathbf{\Pi}_t \circ \left(\mathbf{1}\mathbf{1}' - \mathbf{T}\right) \circ \mathbf{mpk}_t\,(\mathbf{k}_t)\,\mathbf{1}'\right] \mathbf{s}_{t-1}}_{\text{Net Foreign Capital Income} + \text{Undepreciated Capital}}. \end{aligned} \tag{21}$$

This equation describes the dynamics of the balance of payments. On the left-hand side we have consumption and saving. On the right-hand side we have total output and undepreciated capital plus net foreign (capital) income, in three additive terms: the first is income (net of tax) and undepreciated capital earned from worldwide investment by domestic investors; the second is gross capital income paid to investors; the third is tax revenues levied on domestic capital income. Equation (21) says that all consumption and new net saving in excess of production (or equivalently, net imports) are financed by a positive net foreign capital income (generated by a positive net foreign asset position). Conversely, a net foreign capital income has to be balanced by a trade surplus.

Because the matrix of portfolio shares $\mathbf{\Pi}_t$ and the vector of GDPs $\mathbf{y}_t$ are functions of the vector capital stocks ($\mathbf{k}_t$), equation (13) can be rewritten as:

$$\mathbf{k}_t = \mathbf{\Pi}\,(\mathbf{k}_t)\;\mathbf{s}_{t-1}\,(\mathbf{k}_{t-1})\,. \tag{22}$$

Definition. Steady-State Equilibrium. We define a steady-state equilibrium as a vector of wages $\mathbf{w}$, natural resource rental rates $\mathbf{m}$, output $\mathbf{y}$, capital stocks $\mathbf{k}$, wealth $\mathbf{s}$, and portfolio shares $\mathbf{\Pi}$ such that for $\boldsymbol{\zeta} = \mathbf{0}$ the agents' utility and firms' profits are maximized, the markets for factors clear, $\mathbf{k}$ satisfies equation (22) for $\mathbf{k}_{t+1} = \mathbf{k}_t$, and $\mathbf{s}$ satisfies equation (20) for $\mathbf{s}_{t+1} = \mathbf{s}_t$.

Intuitively, at the steady-state equilibrium, all the terms of equation (22) are unchanged with respect to the previous period.

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Dropping the time subscripts, this definition implies that in the steady state:

$$\mathbf{s}_{(\mathbf{k})} = \left\{\boldsymbol{\Theta}^{-1} - \left[\boldsymbol{\Pi}_{(\mathbf{k})} \circ \left(\mathbf{1}\mathbf{1}' - \mathbf{T}\right) \circ \mathbf{mpk}_{(\mathbf{k})}\mathbf{1}'\right]\right.$$
$$\left. - \operatorname{diag}\left[\left(\boldsymbol{\Pi}_{(\mathbf{k})} \circ \mathbb{E}\mathbf{R}_{(\mathbf{k})}\right)'\mathbf{1}\right]\right\}^{-1} (\boldsymbol{\Lambda} + \boldsymbol{\Xi})\,\mathbf{y}_{(\mathbf{k})}. \tag{23}$$

By plugging equation (23) into equation (22), we can solve numerically for the steady-state vector of capital stocks $\mathbf{k}$. Notice that there is a trivial equilibrium at $\mathbf{k} = \mathbf{0}$. When we solve equation (22), we can rule out this trivial equilibrium by solving for the log of $\mathbf{k}$.

It is natural to ask at this point: what happens in the steady-state equilibrium? By definition, bilateral and multilateral assets and liabilities positions are constant. This implies that the current account is also zero. However, this does not mean that the trade balance or net investment positions have to be zero.

The steady-state version of equation (21) allows for permanent capital account imbalances and only requires that the current account be balanced. In the steady state, there will be countries that run a net positive foreign asset position: these countries are able to run a permanent trade deficit because the trade deficit is offset in the current account by a positive stream of net foreign capital income, generated by the positive net foreign asset position. Conversely, there will also be countries that run a permanent negative foreign asset position: these countries also run a permanent trade surplus.

Underlying these steady-state imbalances is a form of comparative advantage: some countries have a comparative advantage in the production of final goods, others in saving (delaying consumption). This leads to gains from trade. Here what is being traded is capital (an intermediate input) in exchange for final goods. In equilibrium, countries automatically sort themselves into "savers" and "producers." Some countries specialize in producing final goods (borrowers/producers), while others act as international financiers, delaying consumption and providing capital in exchange for future consumption. This exchange is mutually beneficial.

II.G. Steady-State Investment Positions and Gravity

To obtain an interpretable expression for the steady-state bilateral investment positions, we can concisely write the

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denominator of equation (14) $\mathcal{M}_j$:

$$\mathcal{M}_j \stackrel{\text{def}}{=} \sum_{\iota=1}^{n} \frac{\mathcal{R}_{\iota j}^{\eta}\ k_{\iota t}}{\Delta_{\iota j}} \tag{24}$$

Then, multiplying both sides of equation (14) by s_j, we can rewrite it with the gross asset position a_{ij} on the left-hand side. These gross positions obey a gravity-like equation:

$$a_{ij} = \frac{\mathcal{R}_{ij}^{\eta}}{\mathcal{M}_j} \cdot \frac{k_i \cdot s_j}{\Delta_{ij}}. \tag{25}$$

Compared with the classical gravity equation in trade, our gravity equation differs in that geographic distance is here replaced by the portfolio wedge Δ_{ij}, and that GDP is replaced by origin country wealth (s_j) and destination country capital stock $k_{\iota t}$.

II.H. Theoretical Results on Steady-State Capital Allocation

We present a series of theoretical results that help us understand under what conditions the competitive steady-state equilibrium of our model produces an efficient allocation of capital and how we can infer allocative inefficiencies from the cross section of rates of return to capital.

We define world GDP (Y) as the sum of the country-level expected outputs:

$$Y_t \stackrel{\text{def}}{=} \sum_{i=1}^{n} y_{it}. \tag{26}$$

Let us call a vector $\mathbf{k} = (k_1, k_2, ..., k_n)'$ a capital allocation. Because labor and natural resources are immobile, Y is a function of $\mathbf{k}$ alone.

Definition 1 (Efficient Capital Allocation). We say that an allocation $\mathbf{k}_t$ is efficient if it maximizes world GDP Y_t given world capital $K_t \stackrel{\text{def}}{=} \sum_{i=1}^{n} k_{it}$, that is:

$$\mathbf{k}_t \in \arg\max_{\mathbf{k}'} Y_t\left(\mathbf{k}'\right) \qquad \text{s.t.} \qquad \sum_{i=1}^{n} k_i' = \sum_{i=1}^{n} k_{it}. \tag{27}$$

The first useful result is that the absence of bilateral frictions produces a CAPM-type environment, where all origin countries hold identical portfolios.

Proposition 1. If asset markets are in equilibrium and there are no bilateral distortions—that is, $\tau_{ij} \equiv \tau_i$ and $\Delta_{ij} \equiv \Delta_i \times$

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Δ_j—then all origin countries j hold identical portfolios of foreign assets (π_{ijt} is independent of j).

Proof. This can be easily verified by substituting $\tau_{ij} \equiv \tau_i$ and $\Delta_{ij} \equiv \Delta_i \times \Delta_j$ into equation (14). Then Δ_j simplifies out from the numerator and the denominator and the resulting expression for π_{ijt} does not depend on j. □

PROPOSITION 2. Consider a steady-state equilibrium: if there are no "objective" distortions—that is, $\tau_{ij} = \Delta_j$ and $\Delta_{ij} = \Delta_j$ (they do not depend on i)— risk-adjusted expected returns are equalized across countries: $\mathcal{R}_i = \mathcal{R}^* \ \forall\, i = 1, 2, ..., n$.

Proof. See Online Appendix B. □

One implication of Propositions 1 and 2 is that, in the absence of distortions, we should observe no home bias. Using this fact, we can proceed to show that equilibrium in input and asset markets implies a direct equivalence between the absence of international frictions and efficient capital allocation. We call this a "dual" efficiency theorem, to emphasize that the effective absence of asset markets frictions translates into factor markets efficiency and vice versa.[9]

THEOREM (DUAL EFFICIENCY). Consider a steady-state equilibrium. The following three statements are equivalent (each is true if and only if the other two statements hold):

(i) Capital is efficiently allocated.
(ii) The marginal product of capital (MPK) is equalized across countries ($\text{MPK}_i = \text{MPK}^*$ for $i = 1, 2, ..., n$).
(iii) Taxes are uniform ($\tau_{ij} \equiv \tau$) and the wedges Δ_{ij} satisfy the following condition:

$$\sum_{j=1}^{n} \frac{\left[\Delta_{ij} \cdot \exp\left(\frac{1}{2}\sigma_i^2 + \sigma_{iw}\right)\right]^{-1} s_j}{\sum_{\iota=1}^{n} \left[\Delta_{\iota j} \cdot \exp\left(\frac{1}{2}\sigma_\iota^2 + \sigma_{\iota w}\right)\right]^{-1} k_\iota} = \mathscr{C} \qquad \text{for } i = 1, 2, ..., n, \tag{28}$$

where $\mathscr{C}$ is some strictly positive constant.

9. This is not a restatement of the first welfare theorem, because it is a statement about GDP, not welfare.

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COROLLARY 1. For a fixed global capital stock K, there is a unique efficient allocation $\mathbf{k}^*$.

COROLLARY 2. Uniform risk premia ($\sigma_i^2 + \sigma_{iw} = \text{constant}$) and frictions ($\Delta_{ij} = \Delta_j$) are jointly sufficient (but not necessary) for statements (i)–(iii) to obtain.

Proof. Online Appendix B. □

Intuitively, the condition outlined in equation (28) requires that risk premia and portfolio frictions offset each other.

III. PORTFOLIO WEDGES: MEASUREMENT STRATEGY AND INTERPRETATION

III.A. Measurement Strategy

To take our model to the data, we must measure the tax rates (τ_{ij}) and the portfolio wedges (Δ_{ij}). To quantify the tax wedges τ_{ij}, we use a direct measurement approach, described in detail in the next section. To measure the portfolio wedge Δ_{ij} we use instead an econometric approach, which we describe here. The guiding principle behind this approach is that, while the portfolio wedges themselves are not directly observable, we observe certain variables that can act as proxies. Our approach is to assume that we can rewrite the wedges as a function of those observables. In what follows, we assume for expositional simplicity that the tax rate does not vary by origin country ($\tau_{ij} = \tau_i$).

For the baseline implementation of the model, we must necessarily focus on a subset of the possible frictions. We choose these specific frictions based on our reading of the previous literature and the plausible exogeneity of the variables used to measure the wedges. Although we deliberately decided to keep the baseline model parsimonious, we believe our model is flexible and amenable to many extensions, so that it will be easy for other researchers to add other frictions. In addition, in Online Appendix H we show how to modify the model to incorporate specific additional frictions, such as capital controls.

We assume that the portfolio wedge can be broken down into two components: one reflects geopolitical distance across countries ($\Delta_{ij}^{\text{Dist}}$); the second reflects political risk in country i

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(Δ_i^{PR}):[10]

$$\Delta_{ij} = \Delta_{ij}^{\mathrm{Dist}} \times \Delta_i^{\mathrm{PR}}. \tag{29}$$

The first portfolio wedge $(\Delta_{ij}^{\mathrm{Dist}})$ reflects cultural and geographic distances among countries. These have been shown to generate "gravity" effects in currency markets (Lustig and Richmond 2020) and in foreign direct and portfolio investment (Portes and Rey 2005; Aggarwal, Kearney, and Lucey 2012; Ahern, Daminelli, and Fracassi 2015). We assume that the log of $\Delta_{ij}^{\mathrm{Dist}}$ can be written as a linear combination of three measures of distance: geographic, cultural, and linguistic:

$$\log \Delta_{ij}^{\mathrm{Dist}} = -\left(\mathrm{GeoDist}_{ij} \cdot \beta_{\mathrm{G}} + \mathrm{CultDist}_{ij} \cdot \beta_{\mathrm{C}} + \mathrm{LingDist}_{ij} \cdot \beta_{\mathrm{L}}\right). \tag{30}$$

Geographic distance is the log of the distance in km between countries i and j; cultural distance is a measure of the dissimilarity in values and beliefs between country i and j, computed based on how individuals from i and j respond to questions from the World Values Survey; linguistic distance is a measure of the expected difference between the language spoken by any two random individuals from countries i and j. We describe all of these variables in detail in Section IV.

To measure the political risk wedge (Δ_i^{PR}), we follow the approach of Alfaro, Kalemli-Ozcan, and Volosovych (2008, AKV) and utilize a composite measure of political risk based on data from the International Country Risk Guide (ICRG), published by the PRS Group. We explain the computation in detail in Section IV.B.[11] We assume that the political risk wedge can be written as a log-linear function of this measure:[12]

$$\log \Delta_i^{\mathrm{PR}} = -\beta_{\mathrm{P}}\, \mathrm{Political\,Risk}_i. \tag{31}$$

10. Due to data limitations, we assume that political risk does not vary at the bilateral level. An interesting extension would be to create a bilateral measure of political closeness PC_{ij} between country i and j.

11. The political risk index is missing for a limited set of countries. We impute their scores using the average of a community of bordering countries that they belong to (e.g., East African Community for Uganda, former Soviet Union for Georgia).

12. In our empirical analysis we use a country-specific measure of political risk rather than a bilateral measure. This limitation means that, in our current implementation, a country's political risk is treated as uniform across all potential investment partners, rather than varying based on bilateral relationships. This is a potential area for future research as more detailed data become available.

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We also assume that in the data, we observe bilateral capital positions a_{ij} with a multiplicative measurement error ε_{ij}:

$$\log \hat{a}_{ij} = \log a_{ij} + \varepsilon_{ij}. \tag{32}$$

By plugging in the expression for the wedges, we can rewrite the gravity equation (25) as:

$$\begin{aligned} \log \hat{a}_{ij} = {} & \alpha_i^d + \alpha_j^o + \text{CultDist}_{ij} \cdot \beta_{\text{C}} + \text{GeoDist}_{ij} \cdot \beta_{\text{G}} \\ & + \text{LingDist}_{ij} \cdot \beta_{\text{L}} + \text{Political Risk}_i \cdot \beta_{\text{P}} + \varepsilon_{ij}, \end{aligned} \tag{33}$$

where the terms α_i^d and α_j^o are equal to:

$$\alpha_i^d \stackrel{\text{def}}{=} \log\left(\mathcal{R}_i^{\eta} k_i\right) \qquad \text{and} \qquad \alpha_j^o \stackrel{\text{def}}{=} \log\left(\frac{s_j}{\mathcal{M}_j}\right). \tag{34}$$

Under the assumption that ε_{ij} is uncorrelated with the variables that proxy for the portfolio wedges, equation (33) represents a linear regression model that can be empirically estimated with country of origin and destination fixed effects. Doing that, we obtain estimates of β_{C}, β_{G}, and β_{L}.

In practice, the fact that taxes are bilateral implies that $\mathcal{R}$ has bilateral variation. In our empirical analysis, we control for bilateral taxes linearly, which is consistent with our theoretical model as a first-order approximation.[13]

The availability of bilateral investment data plays a crucial role in our ability to identify these elasticities: it allows us to control for country of origin (i) and country of destination (j) fixed effects; these multilateral fixed effects capture not only their model-implied counterparts but also any additional country-level confounder or systematic measurement error that is not explicitly modeled. Combined with a rich set of control variables, and the "deep," slow-moving nature of our distance metrics, this reliance on bilateral variation further strengthens our ability to econometrically identify β_{C}, β_{G}, and β_{L}.

It is easy to notice that the regression equation (33) does not allow us to estimate β_{P}, because the variable Political Risk$_i$ only varies by destination country and does not contain bilateral variation; therefore, it will be absorbed by the destination country fixed

13. Using a first-order approximation around ($\tau_{ij} = \overline{\tau}$, $\text{MPK}_i = \overline{\text{MPK}}$) such that $\overline{\tau} \cdot \overline{\text{MPK}} = \delta$, the log risk-adjusted return is approximately:

$$\log \mathcal{R}_{ij} \approx (\tau_{ij} - \bar{\tau})\,\overline{\text{MPK}} + \bar{\tau}\left(\text{MPK}_i - \overline{\text{MPK}}\right) - \frac{1}{2}\sigma_i^2 - \sigma_{iw}. \tag{35}$$

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effect. For this reason, we use a different strategy to obtain an estimate of β_{P}: we use the empirical estimates of AKV, who estimate econometrically the sensitivity of foreign investment inflows (in millions of US$) to this measure of political risk. In Online Appendix D, we explain how we recover the semi-elasticity coefficient β_{P} from AKV's tables.

Equation (33) is our main econometric specification. The dependent variable is measured using data on *Foreign Equity Investment*, *Foreign Debt Investment*, and the sum of the two (*Foreign Assets*).[14] Since the vector of distances varies at the level of the undirected country pair, in our regression analysis we compute standard errors clustered by undirected country pair. Additional bilateral variables are used either as instruments or control variables, depending on the specific empirical model under consideration.

Although some of our measures, such as cultural and geographic distances, are inherently undirected, our model does incorporate directed frictions. Specifically, tax rates (τ_{ij}) and political risk (Δ_i^{PR}) are directed measures that depend on the direction of investment. This allows our model to capture asymmetries in investment frictions between country pairs, reflecting real-world scenarios where barriers to investment may indeed differ depending on the direction of the investment.

III.B. Interpretation of the Distance Wedge

As we clarified already, we use measures of geographic distance, cultural distance, and linguistic distance (alongside political risk) to quantify the portfolio wedges. The interpretation of these measures of geographic and cultural distances, as drivers of international asset demand, is consistent with the previous literature. In particular, a widely prevailing view, backed by the empirical work of Portes and Rey (2005), is that these measures rep-

14. In the Online Appendix, we also consider the determinants of global asset holdings, distinguishing between foreign direct investment and foreign portfolio investment, as is often done in the literature. We prefer to focus on the debt/equity distinction in the main analysis because the distinction between foreign direct investment (FDI) and equity foreign portfolio investment (FPI) is somewhat arbitrary. For a discussion of this point, see Blanchard and Acalin (2016).

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resent information frictions.[15] An alternative view is that these distances are proxies for asset trade costs.

In Online Appendix A, we present alternative microfoundations for our asset demand system. Different interpretations are consistent with such alternative microfoundations. Under the rational-inattention interpretation, these distance measures can be viewed as factors that increase the cost of acquiring information about foreign investments. For instance, greater cultural distance might make it more difficult for investors to understand and interpret information about a foreign market.

Under the second microfoundation, based on extreme-value distributed asset trade costs, we can interpret these distance measures as proxies for systematic trading costs that are incurred by investors when they invest from a specific origin country to a specific destination country. For example, linguistic distance might correlate with the costs of monitoring foreign investments or navigating different legal systems.

Regardless of the specific interpretation, our empirical strategy allows us to quantify the economic importance of these barriers to international investment. By estimating the effect of these distance measures on investment positions, we can calibrate our model and conduct counterfactual analyses that shed light on the potential gains from reducing international investment frictions.

IV. Data

We present the data used in our quantitative analysis, including the bilateral data used in the estimation of the gravity equation for international investment, as well as the country-level variables used to take the model to the data.

IV.A. Newly Developed Data: The Geopolitical Distance Dataset

One of our contributions is to undertake a comprehensive data collection and processing exercise to build a novel,

15. To explore whether these variables do indeed capture information frictions, we looked at their effect on the Social Connectedness Index, which is based on Facebook data (Bailey et al. 2021). We found that cultural, linguistic, and geographic barriers bear an economically and statistically significant relationship with social connectedness. These results are available on request, and a more complete analysis of the relationship between these indicators and the extent of information linkages is left for future research.

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wide-coverage, high-precision database of bilateral measures of international taxation and geopolitical distance between countries. We call it the Geopolitical Distance Data Repository, and we make it publicly available at www.geopoliticaldistance.org. In this subsection, we describe its contents.

1. *Bilateral Capital Tax Rates (Statutory and Effective).* To measure the tax rate τ_{ij}, we build a novel database of bilateral tax rates on capital that covers 225 countries (50,625 country pairs). Our bilateral tax rates are constructed from corporate income tax rates and treaty-adjusted withholding tax (WHT) rates on dividends and interest income from a variety of sources. By "treaty-adjusted," we mean that the applicable WHT rate depends not only on the source of income but also whether the income recipient is domestic or resident in a country that benefits from a double taxation treaty (DTT) with the taxing jurisdiction.

It is important to clarify that these bilateral tax rates should be interpreted as "summary" rates: in reality the national tax codes and the tax treaties contain a multiplicity of provisions that multiply the number of applicable tax rates. We provide details on the construction of the database in Online Appendix C. Here we focus on how we combine the various measures into a composite tax rate on capital.

Denoting the statutory tax rates with the ^{0} superscript, we compute the statutory composite tax rate on capital weighting taxes levied on equity and debt capital with the corresponding equity and debt shares of incoming international investment:

$$\tau_{ij}^0 \stackrel{\text{def}}{=} \frac{\text{Equity}_i}{\text{Equity}_i + \text{Debt}_i}\left(1 - \frac{\text{CIT}_i}{100}\right)\left(1 - \frac{\text{DWHT}_{ij}^0}{100}\right) + \frac{\text{Debt}_i}{\text{Equity}_i + \text{Debt}_i}\left(1 - \frac{\text{IWHT}_{ij}^0}{100}\right), \tag{36}$$

where Equity_i is the foreign equity liability position of i; Debt_i is the foreign debt liability position of i from the datasets of Coppola et al. (2021) and Damgaard, Elkjaer, and Johannesen (2019); DWHT is the dividend WHT rate; and IWHT is the interest WHT rate.

One problem that further complicates the estimation of the applicable tax rate is that issuers can use tax havens to elude taxes, particularly WHTs. As a result, the effective tax rate can deviate substantially from the statutory tax rate (which is why

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we denoted the measure with the ^{0} superscript). In what follows, we derive and propose a measure of the effective bilateral tax rate on capital applicable to investments from j to i.

We start by outlining a simple addendum to our model that endogenizes issuance in tax havens and effective tax rates, without affecting the portfolio shares from the baseline model. We assume that each destination country's representative firm can choose to issue a fraction π_{ijt}^{th} of the capital purchased by investors from j not directly but indirectly through tax havens. Tax havens allow investors to access a tax rate τ_{ijt}^{th} that is lower than the statutory rate τ_{ijt}^{0}.[16] However, issuing securities through tax havens comes with a cost $\mathfrak{c}(\pi_{ijt}^{\text{th}})$, which is expressed as a share of capital income and which we assume to be collected by a competitive intermediary that rebates it back to workers, alongside other government revenues. This cost is increasing and convex in π_{ijt}^{th}.

Thus, for every origin country j, the destination country representative firm i minimizes the effective tax rate on capital, which we denote by τ_{ij}. Formally:

$$\tau_{ij} = \max_{\pi_{ij}^{\text{th}} \in [0,1]} \tau_{ij}^{0} + \left(\tau_{j}^{\text{th}} - \tau_{ij}^{0}\right) \pi_{ij}^{\text{th}} - \mathfrak{c}\left(\pi_{ij}^{\text{th}}\right), \tag{37}$$

where τ_{ij}^{0} is the de jure tax rate and τ_{ij}^{th} is the tax rate that applies to investments that are routed through tax havens. Making the assumption that the cost function $\mathfrak{c}(\pi_{ij}^{\text{th}})$ is quadratic in π_{ij}^{th}

$$\mathfrak{c}\left(\pi_{ij}^{\text{th}}\right) = \frac{1}{2\beta^{\text{th}}} \left(\pi_{ij}^{\text{th}}\right)^2, \tag{38}$$

16. In the specification above, the only reason borrowers in the destination country want to issue shares through tax havens is to avoid (higher) de jure taxes. In reality there is another reason: to avoid regulations and restrictions. For example, in their survey, Florez-Orrego et al. (2024, 645) write: "As discussed in detail in Coppola et al. (2021) and Clayton et al. (2023), this massive adjustment for China arises because of the use by Chinese firms of the variable interest entity (VIE) structure. Chinese regulations forbid firms in sensitive industries, including much of the technology sector, from receiving foreign equity investment. Alibaba, Tencent, Baidu and other Chinese tech firms side-step these restrictions with a VIE structure that simultaneously allows the firms to sell equity claims on a Cayman Island-based shell to foreign investors and tells Chinese regulators that the local companies in China only have domestic equity owners."

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we obtain the following solution to the tax haven issuance problem:

$$\pi_{ij}^{\text{th}} = \min\left[\beta^{\text{th}}\left(\tau_{ij}^{\text{th}} - \tau_{ij}^{0}\right), 1\right]. \tag{39}$$

To compute τ_{ij}^{th}, in equation (36) we replace the WHT rates on dividends and interest with the corresponding statutory WHT rate charged on average by tax havens against the origin country j.[17] This definition is based on the simplifying assumption that issuance through tax havens helps avoid WHTs but not corporate income taxes (we believe this to be a reasonable approximation). Assuming that β^{th} is known, we obtain our estimate of the effective tax rate by plugging this solution into the maximand of equation (37).

The only remaining issue is to estimate β^{th}. A natural guess for this parameter is one, based on the consideration that it equates the share of capital issued in the tax haven to one when the tax rate is equal to 100%. In Section V.C, we estimate this parameter econometrically using a Tobit regression (π_{ij}^{th} is naturally bounded between zero and one). As our estimate of π_{ij}^{th}, we use the difference between nationality- and residency-based positions reported in the datasets of Coppola et al. (2021), Beck et al. (2024), and Damgaard, Elkjaer, and Johannesen (2019). In particular, for every origin-destination country pair, we estimate π_{ij}^{th} (the share issued through tax havens) as the difference between the nationality-based and the residency-based position (if positive) as a ratio of the first. We run the analysis separately for equity and debt. The analysis supports our initial guess $\beta^{\text{th}} = 1$. A final point is that our measure of taxation does not include expropriation risk. This will be measured separately as part of the political risk wedge.

2. *Cultural Distance.* We present a new dataset of cultural distance between countries, which improves on the one previously introduced by Spolaore and Wacziarg (2016). Cultural distance captures distance in contemporary values and beliefs, and it is constructed using response data from the World Values Survey—European Values Survey (WVS-EVS) 1981–2021 Integrated Questionnaire. The questions pertain to the following

17. We use the same definition of tax havens used by Coppola et al. (2021).

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topics: perceptions of life, environment, work, family, politics and society, religion and morale, and national identity.

The new dataset uses a much larger set of questions than the previous one—495 (up from 98)—and it covers a significantly larger set of countries—116 (up from 72). The main reason for this improvement in coverage and number of questions lies in a new methodology that we developed to deal with one of the fundamental issues of the WVS-EVS data—namely, the fact that not all questions are asked in all countries. In the previous iteration of the dataset, this created a stringent trade-off between the number of questions and the number of countries covered (a question could only be used if available for all included countries, and a country could only be included if all relevant questions were asked in the corresponding country survey). Although the previous database (98 questions, 72 countries) was structured to minimize the loss of data, this problem was severe: more than 80% of the data could not be used.

For the new iteration of the dataset, we have devised a method, which we call "flex." It allows (by imputing missing questions) to use a flexible set of questions to estimate the cultural distance between any two given countries (see Online Appendix C for details). The resulting measure of cultural distance is still completely comparable across country pairs. Yet because this method prevents the loss of valuable data, we obtain significant gains in both precision and coverage. Our new measure of cultural distance also uses the most recent version of the WVS-EVS survey, which includes additional countries. This also helped expand the dataset's coverage.

We have also revised the definition of the cultural distance variable so that its scale is now easier to interpret. For two countries A and B, it is now defined as the expected disagreement between two respondents randomly drawn from A and B on any given question used. "Disagreement" is a variable that equals zero if two respondents provide the same answer to the same question and one when they provide different answers. For questions that require an answer on an ordered scale (as opposed to a binary or multinomial answer), disagreement can also take on a value between zero and one—in which case, it will be equal to the absolute value of the difference in the answers, normalized by the size of the scale. Therefore, two countries will have distance zero if the union of their respondents provide the same answers to all of the questions, and will be equal to one if it is not possible

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to find a pair of respondents from those two countries that gives the same answer to at least one question. The new methodology is described in detail in Online Appendix C.

3. *Linguistic Distance.* We present a new database of linguistic distances between countries, which also improves on that previously introduced by Spolaore and Wacziarg (2016). Linguistic distance is based on the fact that different contemporary languages descended from common ancestral languages over time. For instance, German, Italian, and French all descend from a common proto-Indo-European language. In turn, Italian and French descend from more recent common ancestral languages (Romance languages stemming from Latin), while German does not. Thus, Italian and French are more closely related to each other than either one is to German. Intuitively, this is analogous to the concept of relatedness between individuals: two siblings are more closely related to each other than they are to their first cousins, because they share more recent common ancestors (their parents) with each other, while they share more distant ancestors with their first cousins (grandparents) and second cousins (great-grandparents).[18]

We construct linguistic distance using data by Ethnologue, which has built a "family tree" of languages. This family tree, in mathematical terms, is a rooted directed tree graph, where languages, protolanguages, and language families are nodes arranged hierarchically. We provide an updated definition of linguistic distance, based on network theory: for two languages, it is the length of the shortest ancestral path connecting them, divided by the length of the shortest ancestral path that traverses the root (a theoretical "language zero," from which all language families descend). We called this metric the normalized tree distance; it takes on values between zero and one. Two languages have distance one if the root is the least common ancestor and distance zero if they are the same language. One advantage of this new definition is that (unlike the previous measure) it automatically adjusts for the fact that different languages may be

18. The analogy is not perfect because individuals have two parents, while languages typically evolve sequentially from "ancestor" languages. For example, according to Ethnologue the ancestors of the Italian language are, in chronological order: Indo-European, Italic, Romance, Italo-Western, and Italo-Dalmatian.

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more or less distant from the root, thus resulting in a more informative and objective measure.

Aside from the definition, a crucial improvement lies in the quality and extent of the data: whereas the previous measure used language trees matched to ethnic populations provided by Fearon (2003), which only covered 433 languages (167,281 language pairs) and 157 countries (12,246 country pairs), the new data cover 6,737 languages (∼45 million language pairs) and 242 countries (58,564 country pairs).

The linguistic distance between two countries A and B is defined as the expected normalized tree distance of all combination of languages spoken in A and B, weighted by the share of the population that speaks each language in each country. We also provide (geopoliticaldistance.org) two upgraded measure of linguistic proximity based on shared cognates. This measure is similar to the tree-based linguistic distance: it uses language population shares from the Ethnologue. However, it differs by the underlying measure of similarity/distance between languages: we use the frequency of cognate words (words from the two languages that share the same etymological root) as a measure of linguistic similarity. Our updated methodology for these measures is detailed in Online Appendix C. Not surprisingly, languages at a higher linguistic distance from each other also tend to have lower linguistic proximity (the correlation between linguistic distance and linguistic proximity is approximately −0.79). In general, speaking less distant and therefore lexically more proximate languages is likely to facilitate information flows even when people do not speak the same language. For example, an Italian speaker can be expected to communicate more easily with a Spanish speaker than with a Greek speaker.

As mentioned, in our empirical analysis we use both cultural distance and linguistic distance as proxies for information frictions and/or asset trade costs. We expect that these variables should act as barriers to global capital allocation, because they capture related but conceptually and empirically distinct dimensions of differences across societies. Cultural distance is about differences in current values and beliefs, and it is likely to stem from divergence in a vast range of cultural traits that are transmitted vertically (from one generation to the next) and horizontally (across unrelated individuals) over time and space. On the other hand, linguistic distance is about historical divergence in one important but very specific trait, language. In fact, the correlation

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between cultural distance and linguistic distance in our dataset is only 0.231, consistent with the observation that several societies can be quite close to each other culturally while more distant linguistically, or vice versa.

4. *Geographic Distance.* In addition to cultural and linguistic distance, we provide new up-to-date, wide-coverage, and high-precision data on geographic distance between countries. For this part of the database, we build on the methodology of CEPII's GeoDist dataset (Mayer and Zignago 2011). Like GeoDist, we also measure geographic distance between two countries as the population-weighted average of the geodesic distances between cities. The main difference is that we base our calculations on the open source GeoNames database, which is continuously updated (thus, we can provide updated country definitions) and also provides a vast increase in the data available for these calculations. In particular, GeoNames provides a census of all the world's cities/towns with a population above 500. In computing geographic distance, we keep for each country the 50 largest cities (versus 25 in GeoDist), and make sure to cover, for each country, at least half of the population reported in GeoNames. Our methodology for computing geographic distance is detailed in Online Appendix C.

Along with geographic distance, we gathered updated bilateral variables that we use as control variables in our regression analysis: border contiguity, latitudinal distance, and longitudinal distance.

5. *Religious Distance.* Finally, through geopoliticaldistance.org, we provide an updated measure of religious distance, which is also constructed considering the number of nodes in historical trees. In this case, the trees consist of religions grouped in related historical categories. For instance, Near Eastern monotheistic religions are subdivided into Christianity, Islam, and Judaism. These are further divided into finer levels of disaggregation. We obtain data on religious adherence for 237 countries (56,169 dyads), from the World Religion Database (WRD). Hence, the new database covers 40 additional countries (15,557 additional dyads). As we did for the tree-based linguistic distance, we use an updated definition, based on normalized tree distance, as our metric of distance between religions.

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In an extension of our main analysis, we use religious distance as an instrument for cultural distance. That is, we assume that the only way more distant religious histories affects barriers to global capital allocation is through their contemporary effects on differences in values, norms, and attitudes, including different attitudes toward religion and morality, which are captured in our measure of cultural distance based on the WVS. We elaborate on this IV approach in Online Appendix F.

IV.B. Externally Sourced Data

1. *Restated Foreign Investment Data.* We use recently developed data on foreign investment positions (stocks) that account for the existence of tax havens. These tax havens may serve as indirect conduits between origin and destination countries. For instance, the Cayman Islands are often used to transit funds between origin and destination countries in a tax-efficient manner. In recent work, Damgaard, Elkjaer, and Johannesen (2019, DEJ) combined FDI data from the International Monetary Fund (IMF)'s Coordinated Direct Investment Survey (CDIS) and the Organisation for Economic Co-operation and Development's Foreign Direct Investment statistics. They restated the data to account for the fact that some countries act as offshore investment centers. In such countries, there is a high concentration of investment companies that only act as investment vehicles and do not actually engage in productive activities. DEJ used cross-border entity ownership data from Bureau Van Dijk's Orbis to reallocate asset ownership from country of residence of the investment vehicle to the nationality country of the ultimate investor, thereby correcting for artificially inflated numbers pertaining to offshore tax havens. This is the source of our FDI data.

For portfolio investment, our sources are Coppola et al. (2021) (for extra-EMU origin countries) and Beck et al. (2024) (for EMU origin countries). They use data from the IMF's Coordinated Portfolio Investment Survey (CPIS) and restate them to account for the presence of shell companies in tax havens, often used to issue securities. To do so, they use reallocation matrices, based on fund holdings data from Morningstar, to convert international portfolio data from CPIS from a residency basis to a nationality basis. Their Foreign Portfolio Investment (FPI) data are further broken down between debt and equity.

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To obtain a measure of total foreign assets (or foreign total investment, foreign assets), we sum the FPI and FDI series (both are in current international US dollars). Furthermore, we create a series of foreign equity investment by adding up FDI and the equity portion of FPI, and a series for foreign debt investment by isolating the debt portion of the FPI series. Again, all these are measured as stocks (as opposed to flow) variables.

2. *Country Macro Data.* The main source of country-level macroeconomic data is the Penn World Tables (PWT, version 10, Groningen Growth and Development Center 2021). The first variable that we obtain from PWT is country output (y_i), which is measured as GDP at current PPP U.S. dollars. The second is labor input (ℓ_i), which is measured as total employment. From the PWT we also obtain a measure of the stock of reproducible capital (k_i) at current PPP dollars, used only for model validation purposes (our model generates capital stocks endogenously).

From the PWT we also obtain the labor income share of GDP (λ_i). We complement these data, when missing, with estimates from the International Labor Office (ILO) Department of Statistics. The share of natural resources rents as a percent of GDP (ξ_i) is obtained from the World Bank. From the World Bank's Wealth of Nations dataset (World Bank 2018) we obtain estimates of the stock of owned wealth (s_i).[19] Due to mismeasurement and data inconsistency, the accounting identity $\sum_{i=1}^{n} k_i = \sum_{i=1}^{n} s_i$ is only approximately respected by the data. We rescale s_i to make sure that the accounting identity holds.

Finally, we compute the country risk premia using MSCI country indices. Starting from daily index values, we compute the annualized volatility for each country/index in 2017. Where this volatility is missing, we impute missing values using a regression of the log of the volatility on the log of GDP per capita.

Note that these country-level variables are not directly used in our gravity regressions in Section V, where they are absorbed by country fixed effects. However, they are used in calibrating and evaluating our structural model in Sections VI and VII. We present them here to provide a complete overview of our dataset.

19. For a few countries where this value is missing, we estimate this variable by combining the capital stock at current and constant prices from PWT with international investment positions from Lane and Milesi-Ferretti (2001).

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3. *Political Risk.* As hinted earlier, to measure political risk we use a composite index that we build from data from the International Country Risk Guide (ICRG), following AKV's methodology. This database contains a series of metrics that evaluate various political and social attributes to gauge a country's political risk level, which can affect its business and investment environment. This dataset is sold by the PRS Group to investors considering international opportunities (including multinational corporations planning global operations), risk management professionals, policy makers, and researchers.

The dataset includes 11 scores: government stability, socioeconomic conditions, investment profile, internal conflict, external conflicts, corruption, military involvement in politics, religious tensions, law and order, ethnic tensions, democratic accountability, and bureaucracy quality.

Following AKV, we exclude socioeconomic conditions and proceed to create a country-level composite index of political risk (ICRG_i that ranges from 0 to 10) by aggregating all of the other components.

4. *Control Variables.* We use variables called colonial relationship—capturing whether two countries in a pair were ever in a colonizer-colonized relationship, and common colonizer, denoting whether the two countries in a pair ever had a common colonizer.[20] In addition, we construct a bilateral dummy variable, common legal origin, that captures whether i and j's legal systems come from the same legal tradition, based on the taxonomy of La Porta, Lopez-de Silanes, and Shleifer (2008).

We obtain the control dummy variables for hard peg and soft peg (which capture the presence of a fixed exchange rates arrangement) from the dataset of de facto foreign exchange regimes of Harms and Knaze (2021). These dummy variables equal one if the currencies of i and j are linked through (respectively) a hard peg or a soft peg, either directly or indirectly (e.g., the two currencies are pegged to the same currencies). We use 2017 data. We also obtained, from the Electronic Data of Investment Treaties (EDIT), data on bilateral investment treaties, which we code as the dummy variable for investment treaty.

20. The data are from CEPII and can be obtained at http://www.cepii.fr/CEPII/fr/bdd_modele/presentation.asp?id=6.

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To control for trade policy, we obtain data on regional trade agreements (RTAs) and their member countries from Mario Larch's Regional Trade Agreements Database (Egger and Larch 2008). We use bilateral dummy variables representing joint memberships in customs union, free trade agreements, and economic integration agreements as of 2017.

IV.C. Coverage and Summary Statistics

Two distinct samples are used in our analysis. The sample used for the model implementation consists of 96 countries (9,216 dyads), covering 92% of world GDP (based on 2017 data from PWT, version 10.1). After merging investment data and explanatory variables, the sample for the gravity regressions covers up to 96 destination countries and 59 investor countries, for a total of 5,605 dyads. Diagonal observations (origin = destination) are not included. We exclude tax havens and a few additional countries with potentially problematic data.[21] Table I displays summary statistics for the regression sample data.

V. Econometric Analysis

V.A. Least Squares Analysis

We begin by performing an OLS regression of equation (33) for the 2017 cross section. Table II reports the estimates (all regressions are unweighted). Column (1) presents estimation results with the log of total assets (i.e., foreign total investment or foreign assets), as the dependent variable. We find that cultural, geographic, and linguistic distances are statistically and economically significant predictors of bilateral asset positions: the slope coefficients corresponding to these three variables are negative, sizable in magnitude (−11.935, −1.580, and −4.157, respectively) and statistically significant at the 99% confidence level.[22] To get a notion of relative magnitudes, the coefficients can be expressed

21. We use the tax haven list of Coppola et al. (2021). In addition, we exclude Venezuela, Palestine, Kuwait, and Qatar.

22. As a robustness check, we calculated geographic distance using the main financial centers of each country pair, as identified by the Global Financial Centers Index, rather than country centroids, where possible. Regression results using this alternative distance measure are presented in Online Appendix Table G.8. The coefficient estimates are virtually unchanged compared with our baseline specification using geodesic distance.

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TABLE I
SUMMARY STATISTICS

Panel A: Directed (dependent) variables					
	Observations	Mean	St. dev.	Min	Max
Assets national. (US$ mln)	10,980	3,999.0	49,379.9	0.0	2,198,076.5
Debt nationality (US$ mln)	10,980	1,254.4	16,677.8	0.0	1,162,567.4
Equity national. (US$ mln)	10,980	2,744.6	35,606.3	0.0	1,563,972.5
Panel B: Undirected (independent) variables					
	Undir. pairs	Mean	St. dev.	Min	Max
Border	4,656	0.031	0.174	0.000	1.000
Colonial relationship	4,656	0.018	0.134	0.000	1.000
Common colonizer	4,656	0.040	0.196	0.000	1.000
Common legal origin	4,656	0.378	0.485	0.000	1.000
Cultural distance	4,656	0.336	0.024	0.223	0.428
Customs union	4,656	0.070	0.256	0.000	1.000
Econ. integr. agreement	4,656	0.200	0.400	0.000	1.000
Free trade area	4,656	0.218	0.413	0.000	1.000
Geographic distance	4,656	8.505	0.964	3.617	9.880
Hard peg	4,656	0.038	0.191	0.000	1.000
Investment treaty	4,656	0.336	0.472	0.000	1.000
Latitudinal distance	4,656	27.960	22.720	0.186	104.096
Linguistic distance	4,656	0.842	0.174	0.013	1.000
Longitudinal distance	4,656	61.027	47.841	0.175	179.045
Regional trade agreement	4,656	0.378	0.485	0.000	1.000
Religious distance	4,656	0.598	0.213	0.009	0.991
Soft peg	4,656	0.251	0.434	0.000	1.000
Tax rate (effective)	4,656	24.914	7.534	6.700	45.960

as the effect of an increase of one standard deviation in the independent variables in terms of a percentage change in foreign assets ($\%\Delta \mathrm{FA}_{ij} = e^{\beta_x \Delta x_{ij}} - 1$). We find large effects of these barriers: an increase of one standard deviation in cultural distance (0.024 units) is associated with a 24.9% decrease in foreign assets, an increase of one standard deviation in geographic distance (0.964 units) is associated with a 78.2% decrease in foreign assets, while an increase of one standard deviation in linguistic distance (0.174 units) is associated with a 51.5% decrease in foreign assets.

In column (2) we present estimation results using log foreign equity investment as the dependent variable. We find again that all three barriers have a statistically and economically significant effect on international equity investment: an increase of one standard deviation in cultural distance (0.024 units) is associated with a 29.0% decrease in foreign equity, an increase of one standard

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TABLE II
OLS REGRESSIONS

Dep. variable in logs	Assets (1)	Equity (2)	Debt (3)	Assets (4)	Equity (5)	Debt (6)
Cultural distance	−11.935**	−14.275**	−13.740**	−14.024**	−16.901**	−13.914**
	(2.813)	(3.560)	(2.898)	(2.998)	(3.742)	(3.202)
Geographic distance	−1.580**	−1.669**	−1.203**	−1.624**	−1.820**	−1.433**
	(0.072)	(0.086)	(0.086)	(0.196)	(0.204)	(0.210)
Linguistic distance	−4.157**	−4.631**	−1.205*	−3.672**	−4.009**	−1.256*
	(0.426)	(0.516)	(0.473)	(0.451)	(0.546)	(0.519)
Observations	4,455	3,948	3,568	4,455	3,948	3,568
R^2	0.733	0.700	0.789	0.744	0.711	0.795
Within R^2	0.241	0.237	0.130	0.271	0.266	0.152
Control variables	No	No	No	Yes	Yes	Yes

Notes. This table reports OLS estimates of a linear regression of the log of the variable listed in the top row (foreign assets, foreign equity assets, foreign debt assets) on the variables in the leftmost column, using data from 2017. Each observation is a directed country pair. All regressions include origin country (i) fixed effects and destination country (j) fixed effects. Additional controls in columns (4)–(6) are border contiguity, latitudinal distance, longitudinal distance, colonial relationship, common colonizer, common legal origin, hard peg, soft peg, customs union, economic integration agreement, free-trade agreement, investment treaty, regional trade agreement, and tax rate (effective). Standard errors (clustered by undirected country pair) are in parentheses. * $p < .1$; ** $p < .01$.

deviation in geographic distance (0.964 units) is associated with an 80.0% decrease in foreign equity, while an increase of one standard deviation in linguistic distance (0.174 units) is associated with a 55.3% decrease in foreign equity. The standardized effects as defined above are slightly larger than those for log foreign assets. Column (3) considers log foreign debt investment as the dependent variable. We find effects of cultural distance (with a standardized effect of −28.1%), geographic distance (a standardized effect of −68.6%), and linguistic distance (with a standardized effect of −18.9%) that are all statistically significant at the 1% level, except for the effect of linguistic distance on debt (significant at the 5% level). These numbers are commensurate with the effects on log foreign assets.

Finally, columns (4)–(6) repeat the analysis of the first three columns but depart from our parsimonious specification by adding controls for a variety of geographic variables (border contiguity, latitudinal distance, longitudinal distance), common history variables (past colonial relationship, common colonizer, common legal origins), as well as variables possibly capturing bilateral facilitators of capital exchange (currency peg, customs union, economic integration agreement, free-trade agreement,

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investment treaty, regional trade agreement, and the bilateral effective tax rate). The goal is to address the possibility that omitted-variables bias affected our main coefficients of interest. The coefficient estimates on cultural, linguistic, and geographic distances are similar in magnitude to those in the parsimonious specification of columns (1)–(3): for foreign assets, we find standardized effects of cultural distance, geographic distance, and linguistic distance to be equal, respectively, to −28.6%, −79.1%, and −47.2%. We again find that these barriers have similar quantitative effects on foreign equity investments and foreign debt investment, though linguistic distance appears to have a smaller effect on foreign debt investment.[23] Overall, adding control variables does not fundamentally alter the inferences drawn from the more parsimonious specification.[24]

V.B. Pseudo-Poisson Regressions

One limitation of the econometric model described by equation (33) is that, being written in logs, it can only accommodate strictly positive capital positions ($\hat{a}_{ij} > 0$), whereas our data contains many zero investment positions. To address this issue and incorporate country pairs with zero investment, we use an alternative specification of our gravity equation.

Following the approach proposed by Santos Silva and Tenreyro (2006) and further developed by Correia, Guimarães, and Zylkin (2020, CGZ), we write an alternative specification of

23. We find somewhat weaker effects of linguistic distance on debt compared with equity, although the effects are still strong and significant. This finding is consistent with previous research, such as Lane (2006), who found that debt flows are less sensitive to gravity variables (proxying for information frictions) compared with equity flows. Lane argued that this difference may arise because debt contracts are simpler and rely more on publicly available macroeconomic information, while equity investments require more detailed, country-specific knowledge.

24. The estimates on the distance variables are also robust to directly including a measure of goods trade flows on the right-hand side of the specification (estimates are available on request). The magnitudes of the semi-elasticities become somewhat smaller, due to the collinearity between trade and distance, but the distance measures remain statistically significant at the 1% level after the inclusion of goods trade. This robustness is consistent with findings in previous research, such as Aviat and Coeurdacier (2007) and Lane and Milesi-Ferretti (2008), who have explored the relationship between bilateral international investment and bilateral international trade. Trade in goods and foreign asset holdings are simultaneously determined, however, so we exclude goods trade from our baseline specification due to endogeneity concerns.

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TABLE III
PSEUDO-POISSON REGRESSIONS

Dependent variable	Portfolio shares					
	Assets (1)	Equity (2)	Debt (3)	Assets (4)	Equity (5)	Debt (6)
Cultural distance	−17.432**	−10.467*	−15.905**	−16.492**	−12.049*	−12.886**
	(3.598)	(4.885)	(3.627)	(3.901)	(5.046)	(4.345)
Geographic distance	−1.371**	−1.630**	−1.105**	−1.287**	−1.663**	−1.234**
	(0.058)	(0.070)	(0.076)	(0.111)	(0.146)	(0.130)
Linguistic distance	−3.353**	−4.104**	−2.521**	−3.474**	−4.145**	−2.647**
	(0.532)	(0.666)	(0.603)	(0.486)	(0.603)	(0.595)
Observations	5,605	5,605	5,546	5,605	5,605	5,546
Control variables	No	No	No	Yes	Yes	Yes

Notes. This table reports iteratively reweighted least squares (IRLS) estimates of a pseudo-Poisson regression of the portfolio shares for each asset class listed on the second row (foreign assets, foreign equity assets, foreign debt assets) on the variables in the leftmost column. Each observation is a directed country pair. All regressions include origin country (i) fixed effects and destination country (j) fixed effects. Additional controls in columns (4)–(6) are border contiguity, latitudinal distance, longitudinal distance, colonial relationship, common colonizer, common legal origin, hard peg, soft peg, customs union, economic integration agreement, free-trade agreement, investment treaty, regional trade agreement, and tax rate (effective). Observations are weighted by the inverse of the destination country GDP. Standard errors (clustered by undirected country pair) are in parentheses. * $p < .1$; ** $p < .01$.

regression equation (33) as:

$$\hat{a}_{ij} = \exp\Big(\alpha_i^d + \alpha_j^o + \text{GeoDist}_{ij} \cdot \beta_{\text{G}} + \text{CultDist}_{ij} \cdot \beta_{\text{C}} + \text{LingDist}_{ij} \cdot \beta_{\text{L}}\Big) + \varepsilon_{ij}. \tag{40}$$

This Poisson pseudo-maximum likelihood (PPML) specification allows for the inclusion of zero observations by modeling the error term as additive rather than multiplicative, a now standard approach in the international trade and finance literature for estimating gravity equations in the presence of zeros. With this method, we can include observations with zero investment positions in our analysis, increasing our sample size and potentially improving the precision of our estimates. To estimate this model, we use the ppmlhdfe Stata command developed by CGZ, which efficiently handles high-dimensional fixed effects in Poisson regressions. For efficiency, we run this equation on portfolio shares and weight observations by GDP of the destination country.

The results of this estimation are presented in Table III. The sample size for total assets rises from 4,455 to 5,605 observations (an increase of about 25.8%), while for equity and debt, it

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increases from 3,948 and 3,568 to 5,605 and 5,546 observations, respectively (increases of about 42.0% and 55.4%). In general, we find that the standardized magnitude of Poisson estimates on cultural, geographic, and linguistic distances are very similar to the corresponding OLS estimates, and the effects remain economically meaningful and statistically significant. For instance, in the specification of column (1), the standardized effect of cultural distance is to reduce total foreign assets by 34.2%, whereas that of geographic distance and linguistic distance are −73.3% and −44.2%. For comparison, unweighted PPML estimates are presented in Online Appendix Table G.7.

Columns (4)–(6) present results with the full set of control variables. The effects of cultural, geographic, and linguistic distance remain robust to the inclusion of these controls, with only modest changes in the magnitude of the coefficients. For foreign assets, the standardized effects of cultural, geographic, and linguistic distances are −32.7%, −71.1%, and −45.4%, respectively.

Broadly speaking, a consideration of the extensive margin does not greatly affect our basic finding that geographic, linguistic, and cultural barriers exert quantitatively meaningful and statistically significant negative effects on foreign asset holdings. The consistency of results across OLS and Poisson specifications lends further credence to the robustness of our findings.

V.C. Tax Haven Issuance Analysis (Tobit)

To analyze the determinants of tax haven issuance, we use a Tobit regression model. This approach is appropriate given that our dependent variable—the share of bilateral investment routed through tax havens—is naturally bounded between zero and one. Our analysis is based on equation (39), with the only modification that we estimate separate models for equity and debt positions: therefore we have separate variables for the share of issuance routed through tax havens as well as the tax savings.

The key independent variable in our analysis is the estimated tax saving from issuing in tax havens. For equity, this is calculated as the difference between the statutory composite tax rate on equity and the tax haven composite tax rate on equity. For debt, it is the difference between the statutory withholding tax rate on interest income and the tax haven withholding tax rate on interest income. Table IV presents the results of our Tobit regressions. Columns (1)–(3) show results for equity positions, and

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TABLE IV
TAX HAVEN ISSUANCE—TOBIT REGRESSIONS

	Issuance share					
Dep. variable	Equity (1)	Equity (2)	Equity (3)	Debt (4)	Debt (5)	Debt (6)
Tax saving	0.513**	0.417*	1.004**	1.777**	1.337**	1.001**
	(0.139)	(0.168)	(0.164)	(0.141)	(0.203)	(0.173)
Issuer fixed effects	No	Yes	Yes	No	Yes	Yes
Investor fixed effects	No	No	Yes	No	No	Yes
Observations	8,754	8,754	8,754	7,592	7,592	7,592
Pseudo R^2	0.001	0.211	0.349	0.010	0.181	0.378

Notes. This table reports OLS estimates of a Tobit regression (with censoring at zero and one) where the dependent variable is the estimated share of j to i investment that is routed through tax havens; the independent variable is the estimated tax saving from issuing in tax havens, measured as the difference in the applicable tax rate. It differs for equity and debt positions. The first three columns present results for equity; the latter three for debt. * $p < .1$; ** $p < .01$.

columns (4)–(6) show results for debt positions. For each type of asset, we present three specifications: a baseline model, a model with issuer fixed effects, and a model with both issuer and investor fixed effects.

For equity, we find a positive and statistically significant relationship between tax savings and the share of investment routed through tax havens. The coefficient on tax savings ranges from 0.417 to 1.004, depending on the specification. This suggests that a 1 percentage point increase in potential tax savings is associated with a 0.4–1 percentage point increase in the share of equity investment routed through tax havens. The results for debt are at least as strong. The coefficient on tax savings ranges from 1.001 to 1.777. A 1 percentage point increase in potential tax savings on interest income is associated with a 1–1.8 percentage point increase in the share of debt investment routed through tax havens.

Including issuer and investor fixed effects helps control for unobserved country-specific factors that might influence the use of tax havens. The persistence of statistically significant coefficients across all specifications suggests that the relationship between tax savings and tax haven usage is robust. Moreover, the coefficient for both equity and debt approaches one when we include additional fixed effects. The pseudo R^2 values increase substantially when we include fixed effects, particularly for the models with both issuer and investor fixed effects. This indicates that country-specific factors explain a significant portion of the variation in tax haven usage.

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Importantly, the slope coefficients from these Tobit regressions are used to compute the βth coefficient in our model. Specifically, we calibrate $\beta^{th} = 1$, based on the most comprehensive specification (columns (3) and (6)) for equity and debt securities, respectively. These β^{th} values are then used to calculate the effective tax rates, which are subsequently used in our quantified general equilibrium model of international capital allocation.

Overall, these results provide strong evidence that tax considerations play a crucial role in determining the extent to which cross-border investment is routed through tax havens. The effect appears to be particularly pronounced for debt positions, which may reflect the greater flexibility in structuring debt instruments for tax purposes. By incorporating these empirically estimated relationships into our broader model, we can more accurately capture the effect of tax haven usage on global capital allocation.

VI. Model Calibration, Fit, and Predictions

Here we calibrate the model of Section II using the econometric estimates of Section V and evaluate how the calibrated model fits the data.

VI.A. Calibrating the Elasticity Parameter η

The parameter η governs the elasticity of substitution among different countries' assets and is therefore an important determinant of the representative investors' portfolios. We calibrate $\eta = 18.3$ based on the previous literature and the fact that for a small open economy i, the elasticity of investment with respect to the expected return is approximately equal to η.

Koijen and Yogo (2020) estimate a demand system for international assets for 2002–2017 and report demand-price elasticities of 3.1 for long-term debt, 25.2 for short-term debt, and 1.2 for equity. To convert the debt price elasticities into gross return elasticities, we multiply by the duration (we assume five years for long term and six months for short term), thus obtaining an average elasticity for debt securities of 15.5 and 12.6 (average = 14.05). To convert the elasticity for equity into a price-gross return elasticity, we can use a back-of-the-envelope calculation based on Gordon's constant dividend growth model: we multiply this elasticity by the gross return and divide it by the difference between the net return and the dividend growth rate. Using the

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average MSCI world return (9.3%) and a dividend growth rate of 2.9% (equal to the world GDP growth over the period), we obtain an elasticity of 20.5. Taking a weighted average that reflects the weight of these asset classes in global portfolios (two-thirds equity, one-third debt), we obtain our calibrated value $\eta \approx 18.3$.

VI.B. Model Solution and Identification

We calibrate the distance semi-elasticities (β) using the estimates of Table II, column (4) (which includes the full set of controls): -14.024 for cultural distance, -1.624 for geographic distance, and -3.672 for linguistic distance. We chose this specification because the magnitude of the effect of the main barrier variables tends to be smaller than in the specifications without controls, or the specifications that use IV estimation (in other words, we choose conservative estimates).[25] We compute the matrix of wedges (Δ_{ij}) using equations (30) and (31).

Armed with empirical estimates for $\boldsymbol{\beta}$ and for a given calibrated value of η, we solve the model. The model takes as inputs (matches) observed GDP, factor shares, and the wealth stock of each country. We solve numerically for the steady-state equilibrium using the key equation: $\mathbf{k} = \mathbf{\Pi}(\mathbf{k})\mathbf{s}$, where $\mathbf{k}$ is the vector of capital stocks, $\mathbf{\Pi}(\mathbf{k})$ is the matrix of portfolio shares (which depends on $\mathbf{k}$ through the risk-adjusted rates of return), and $\mathbf{s}$ is the (observed) vector of wealth. This solution gives us the predicted equilibrium portfolio shares, capital stocks, and therefore rates of return for each country.

Armed with the vector of capital stocks, we rewrite the Cobb-Douglas production function of country i by grouping nonmobile factors (including technology) in one single term $\hat{\omega}_i$:

$$y_i = \hat{\omega}_i k_i^{\kappa_i}, \tag{41}$$

where

$$\hat{\omega}_i \stackrel{\text{def}}{=} \omega_i x_i^{\xi_i} \ell_i^{\lambda_i}. \tag{42}$$

We compute country-level saving rates (θ_j) from (observed) output, returns, and wealth using equation (17). The last model component that remains to be identified is $\hat{\omega}_i$: this is obtained from equation (41).

25. In Online Appendix G.7 we examine the sensitivity of the counterfactual analysis to the use of alternative estimates of β, finding that such alternatives deliver broadly similar results to those of the benchmark exercise.

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VI.C. Model Fit

1. *Portfolio Shares.* To evaluate the model's empirical performance, we compare untargeted data moments generated by the model against their empirical counterparts. As observed by Armenter and Koren (2014), some data moments are less informative than others when it comes to evaluating model fit, as they can be reproduced equally well by a rudimentary/mechanical model, and therefore (likely) by a large set of alternative models. In the case of our model, it is important to understand to what extent our matching of data moments is due to the presence of frictions, as opposed to other features of the model.

To this end, we produce a variant of the model to serve as a benchmark in evaluating model fit. This benchmark, which we call the "frictionless" model, is identical to the baseline model except that all frictions and taxes are removed (i.e., $\Delta_{ij} = 1$ and $\tau_{ij} = 1$ for all i, j). In this frictionless model, neither information frictions, policy barriers, nor taxes play a role in capital allocation. Under this benchmark, each origin country invests in other countries solely based on their risk-adjusted returns, and their portfolios are all identical (see Proposition 1).

We begin evaluating our model by looking at how well it can fit the empirical external portfolio shares (π^e_{ij}). These external portfolio shares are defined as the share of country j's foreign investment allocated to country i, excluding domestic investment:

$$\pi^e_{ij} = \frac{a_{ij}}{\sum_{\iota \neq j} a_{\iota j}} \qquad i \neq j. \tag{43}$$

We use external portfolio shares because we do not observe domestic investment in the IMF-derived data.

In Figure II, we compare the model-implied external portfolio shares to the actual ones, computed using IMF restated data. The figure plots the model-implied shares against the actual shares, with each point representing a country pair. The upper panel shows the comparison for the baseline model, and the lower panel shows it for the frictionless benchmark. The baseline model reproduces the general pattern of external portfolio shares observed in the data, with a mean squared error (MSE) of 1.11. Although this indicates some deviation from the actual data, it represents a reasonably good fit considering the model's parsimony. In contrast, the frictionless model performs notably worse, with an MSE of 1.76. This higher MSE indicates that the frictionless model's predictions deviate more substantially from the observed data.

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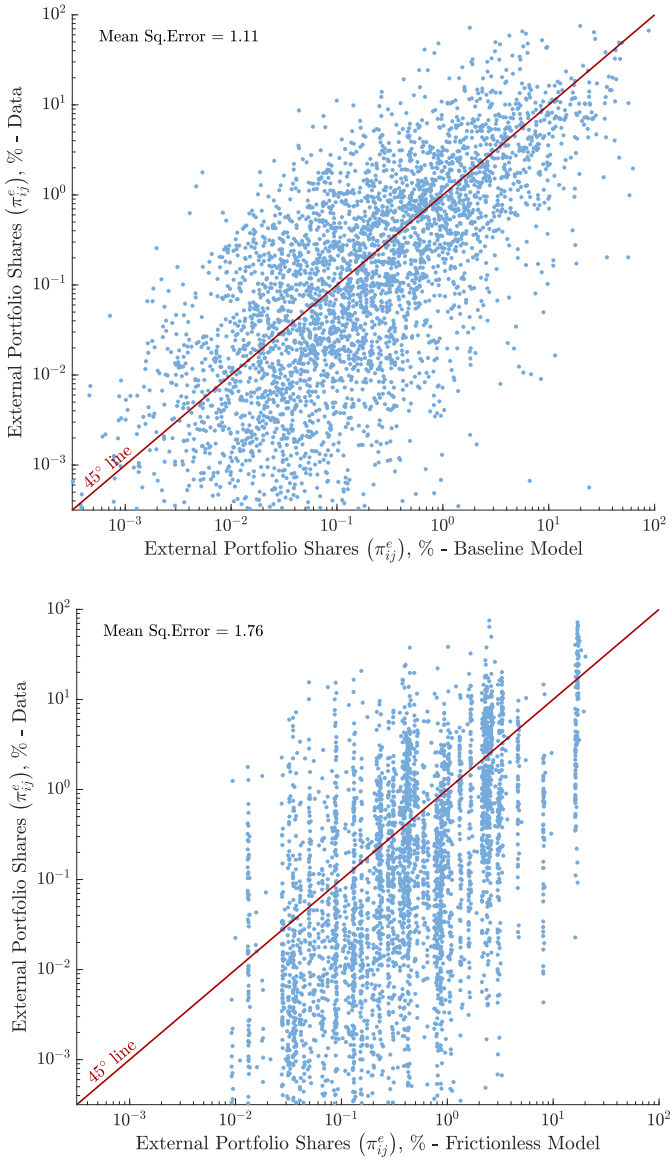

FIGURE II

Model Fit: External Portfolio Shares

The figure plots the model-implied external portfolio shares (π^e_{ij}), in percentage, against the actual ones, computed using IMF restated data. Every dot is a country pair. The model portfolio shares of the top panel are from the baseline, while those from the bottom panel are from the benchmark "frictionless" model.

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The lower panel of Figure II shows a pattern of vertical lines for the frictionless model, which is expected: by Proposition 1, a frictionless model produces portfolio shares that are symmetric across origin countries. Each vertical line corresponds to a specific destination country. The bilateral frictions in the baseline model crucially allow us to break this symmetry and thus fit the country portfolios more satisfactorily, as evidenced by the lower MSE.

2. *Nationality Versus Residency Positions.* To further validate our model's performance and address a key issue from earlier work, we examine how well it predicts nationality-based versus residency-based portfolio positions. Figure II shows the model fit for nationality-based external portfolio shares, while Online Appendix Figure E.1 presents the corresponding plot for residency-based shares. Comparing these figures reveals that our model provides a better fit for nationality-based positions.

The nationality-based plot (Figure II) shows a tighter clustering of points around the 45-degree line, indicating a closer match between model predictions and actual data. In contrast, the residency-based plot (Online Appendix Figure E.1) exhibits more dispersion, with many points deviating substantially from the line of perfect fit. This improvement in fit is quantified by the difference in MSE between the two models: the MSE for residency-based positions is 1.23, while for nationality-based positions it decreases to 1.11. This superior fit for nationality-based positions is particularly noteworthy given that our model was not explicitly designed or calibrated to capture the distinction between nationality-based and residency-based investment. Rather, this emerges as a natural consequence of the model's structure, which incorporates frictions that can lead to the use of offshore financial centers and other intermediaries in cross-border investment.

The model's ability to better predict nationality-based positions aligns with recent empirical work by Coppola et al. (2021), highlighting the importance of looking beyond immediate counterparties to ultimate ownership in international investment. It suggests that our framework captures key economic forces driving global capital allocation, including the incentives for indirect investment routes that can obscure true bilateral positions in residency-based data. This result validates our modeling ap-

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TABLE V
MODEL FIT: UNTARGETED MOMENTS

Variable	Statistic	Data	Model: Baseline	Model: Frictionless
Return to capital	Mean	−2.121	−2.081	−2.246
$\log(\mathrm{MPK}_i)$	Standard deviation	0.496	0.422	0.091
	Correlation w/ data	1.000	0.663	0.319
Capital/employee	Mean	11.811	11.771	11.936
$\log(k_i/\ell_i)$	Standard deviation	1.147	1.192	0.917
	Correlation w/ data	1.000	0.947	0.918
Home bias	Mean	4.006	4.000	0.000
$\log \pi_{ii} - \log \frac{k_i}{K}$	Standard deviation	1.224	1.078	0.000
	Correlation w/ data	1.000	0.945	0.000

proach and underscores the potential of our framework for analyzing discrepancies between nationality-based and residency-based investment patterns.

3. *Untargeted Moments.* Table V presents moments of the data, against the corresponding model-generated moments for our baseline model and the frictionless benchmark. We look at three different key variables: physical rates of return on capital (MPK_i), capital stock per employee ($\frac{k_i}{\ell_i}$), and home bias, which we define (following Lau, Ng, and Zhang 2010) as

$$\text{Home Bias}_i \stackrel{\text{def}}{=} \log \pi_{ii} - \log \frac{k_i}{K}. \tag{44}$$

Our sources for the Data column are as follows. Rates of return on capital are computed using the methodology of Monge-Naranjo, Sánchez, and Santaeulàlia-Llopis (2019). This computation requires output, capital stock, and labor shares from the PWT as well as natural resource shares from the World Bank Wealth of Nations dataset. For capital stock per employee, we use the corresponding data from the PWT. To compute home bias, we use the estimates of Lau, Ng, and Zhang (2010) of the percentage of local funds' holdings in domestic securities as the estimate for π_{ii}, and PWT estimates of $\frac{k_i}{K}$.

Overall, the baseline model comes closest to matching the data. For rates of return on capital, the frictionless model produces much less variation (standard deviation of 0.091) compared with the data (0.496). In the frictionless model, the only source of

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variation in MPK is risk, which explains some of the observed patterns but is insufficient to match the data. The baseline model, by incorporating frictions and taxes, comes much closer to matching the dispersion in returns (standard deviation of 0.422). Moreover, the baseline model's rates of return correlate more closely with the data (correlation of 0.663) than those from the frictionless model (correlation of 0.319).

For capital stock per employee, the baseline model slightly overpredicts the dispersion (standard deviation of 1.192 compared with 1.147 in the data), while the frictionless model underpredicts it (0.917). Both models show high correlations with the data, but the baseline model performs marginally better (0.947 versus 0.918).

As implied by our theory, the frictionless model does not generate any home bias, while the baseline model produces substantial home bias. The baseline model matches the data closely in both average magnitude (4.000 versus 4.006) and cross-sectional dispersion (1.078 versus 1.224). Figure III provides a visual representation of how well our model captures home bias. It plots the model-implied home bias against the empirical home bias for each country. The strong positive relationship evident in this figure is quantified by the very high correlation of 0.945 between the model-generated home bias and the data.

Given that our gravity equation loads negatively on measures of cultural and geographic distance and incorporates tax differentials, the model's prediction of home bias is not entirely surprising. What is noteworthy is the model's ability to match not only the overall level of home bias but also the specific values for individual countries with such high accuracy. This is clearly demonstrated by the tight clustering of points along the 45-degree line in Figure III. The fact that our model can reproduce the pattern of home bias so accurately for the vast majority of countries, despite not being explicitly calibrated to do so, provides strong validation for our approach to modeling international capital allocation.

In summary, the baseline model's superior performance in matching these untargeted moments, particularly compared with the frictionless benchmark, underscores the importance of the frictions and taxes we have incorporated for explaining patterns of international capital allocation. The improvement in fit from the frictionless model (MSE 1.76) to the baseline model (MSE 1.11) for external portfolio shares, combined with the high

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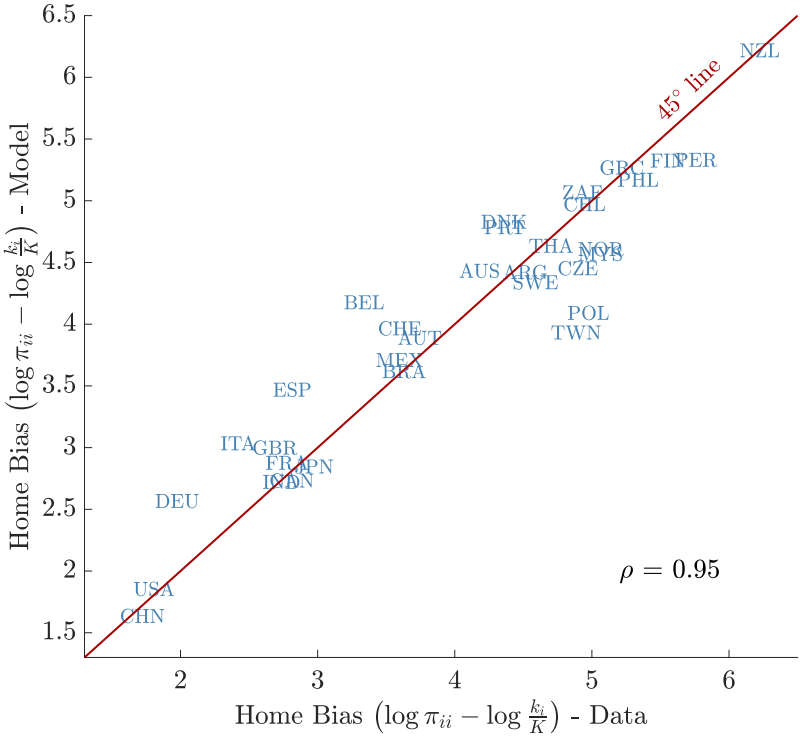

FIGURE III

Model Fit—Home Bias (Untargeted)

This figure plots the model-implied home bias, against the empirical counterpart, which is computed by combining data from Lau, Ng, and Zhang (2010) with the Penn World Tables. Each observation is a country.

accuracy in predicting home bias, further demonstrates the role of these factors in capturing real-world investment patterns.

4. *Rates of Return Heterogeneity.* In addition to matching data moments well, our model replicates some stylized facts that the literature has documented. As noted by David, Henriksen, and Simonovska (2014, DHS), the marginal product of capital (MPK) correlates negatively, at the country level, with economic development. In Figure IV, we plot the relationship between the (log) MPK from our model against the log of GDP per employee. The correlation between these two variables is −0.7: this is

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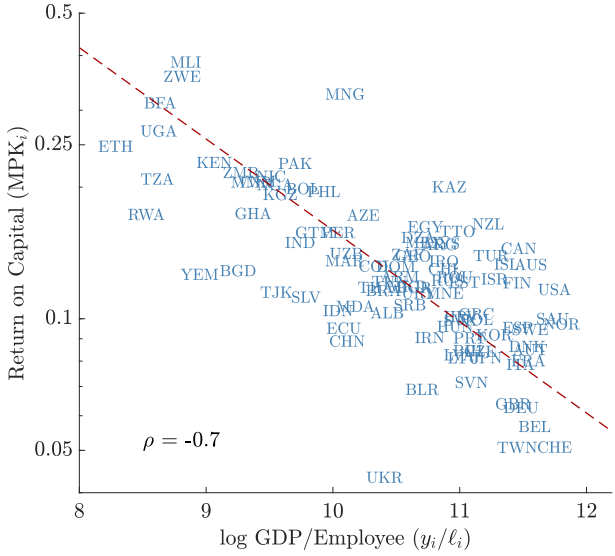

FIGURE IV

Country Heterogeneity in Returns to Capital

This figure plots the model-implied rate of return on capital against the log of GDP per employee. Each observation is a country and the data are for 2017.

consistent with DHS's observation that the marginal product of capital is significantly higher in emerging economies.

If movements of capital were unimpeded, we would expect capital to migrate from richer to poorer countries to reduce these return differentials. Return differentials are a reflection of Lucas's observation about the paucity of such flows in the data. In our model, these return differentials are produced by differences in the risk properties of countries as well as by our measured wedges. Consequently, these barriers help explain the absence of large movements of capital toward developing countries, shedding light on the Lucas puzzle.[26]

26. DHS also develop a model to explain this stylized fact. In their theoretical framework, capital yields higher returns in emerging economies due to risk and diversification (emerging assets are a worse hedge for global risk). In our baseline model, returns to capital are higher in emerging markets not only due to their risk properties but also due to frictions.

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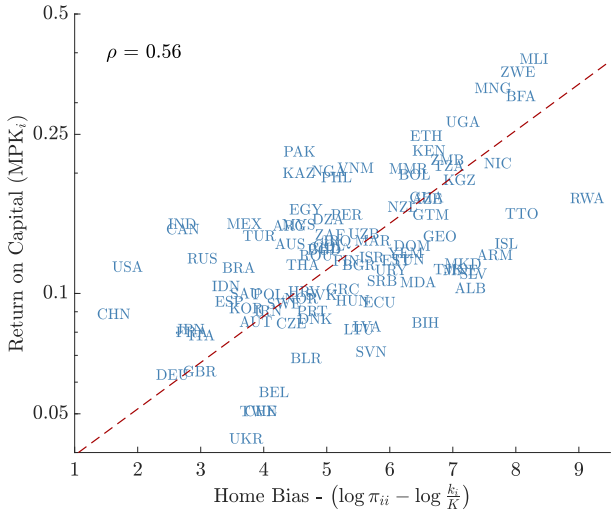

FIGURE V

Rates of Return and Home Bias

This figure plots the model-implied rate of return on capital against a measure of home bias—the log difference of the shares of country i investment in country i's portfolio and in the world portfolio ($\log \pi_{ii} - \log \frac{k_i}{K}$). Each observation is a country and data refer to the year 2017.

5. *Home Bias and Rates of Return.* Another stylized fact that our model is able to account for is that home bias correlates positively with rates of return. This fact was robustly documented by Lau, Ng, and Zhang (2010). To show that our general equilibrium model is capable of reproducing this correlation, we compute our own model-consistent version of this measure (equation (44)) and plot it against the model-implied MPK in Figure V. As seen in the graph, the two correlate strongly and positively ($\rho = 0.56$).

This relationship arises due to the fact that countries with higher home bias face difficulties accessing international assets. Simultaneously, international investors demand a return premium to invest in these less accessible countries, leading to higher rates of return on capital.

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VII. Counterfactual Analysis

VII.A. Capital Allocation Efficiency

If we could exogenously change the set of barriers affecting international investment and let market forces reallocate capital, how would the cross-country distribution of capital and output change? What would the efficiency gains be? In this section, we explore this question using counterfactual analysis. Our counterfactuals consist of selectively removing the various wedges within our model (taxes and portfolio wedges). To remove a wedge, we change the (previously estimated) matrix of wedges to a matrix of ones. For each of the counterfactuals, we compute the corresponding world GDP. We also compute the percentage difference between the counterfactual and an undistorted (frictionless) equilibrium in terms of three statistics: world GDP, the standard deviation of the log of capital per employee, and the standard deviation of log of output per employee.[27]

Table VI presents the main results from the counterfactual analysis. In column (1), we present the observed equilibrium which is distorted by geopolitical distance, taxes, and political risk. In column (2), we present the frictionless equilibrium, from which all distortions have been removed ($\Delta_{ij} = 1\ \forall i, j$). In column (3), we consider a counterfactual equilibrium (taxes only) where distortions from both political risk and geopolitical distance are eliminated ($\Delta_{ij}^{\text{Dist}} = \Delta_i^{\text{PR}} = 1\ \forall i, j$) while distortions from taxes remain in place. In column (4), we consider a counterfactual equilibrium (political risk only) where distortions from taxes and geopolitical distance are eliminated ($\Delta_{ij}^{\text{Dist}} = \tau_{ij} = 1\ \forall i, j$) while distortions from political risk remain in place. In column (5), we consider a counterfactual equilibrium (geopolitical distances only) where distortions from taxes and political risk are eliminated ($\tau_{ij} = \Delta_i^{\text{PR}} = 1\ \forall i, j$) while distortions from geopolitical distance remain in place. These three counterfactuals allow us to gain a sense of the marginal effect of each individual distortion.

We find that barriers to the global allocation of capital have quantitatively important effects on the level of output produced globally. World GDP in the observed equilibrium of our model is measured at US$112.9 billion. That is 7.0% lower than in the frictionless counterfactual (column (2)). We find that distortions from geopolitical distance have the largest effect in terms of cap-

27. Our baseline counterfactual analysis will focus on steady-state outcomes. An extension to out-of-steady-state outcomes is provided in Online Appendix G.1.

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TABLE VI

COUNTERFACTUALS (2017)

Welfare statistics	Observed (all barriers) (1)	Frictionless (no barriers) (2)	Taxes only (3)	Political risk only (4)	Geopolitical distance only (5)
World GDP (PPP$ trillions)	112.9	121.4	118.0	120.8	114.8
World GDP, % difference from (2)	−7.0%	+0.0%	−2.8%	−0.4%	−5.4%
St. dev. of $\log(\frac{k_i}{\ell_i})$, % difference from (2)	+50.9%	+0.0%	+13.0%	+7.4%	+38.3%
St. dev. of $\log(\frac{y_i}{\ell_i})$, % difference from (2)	+22.0%	+0.0%	+5.9%	+3.9%	+15.7%

Notes. This table presents welfare statistics for four counterfactuals of the model described in Section II. Each of columns (2)–(5) is a counterfactual, and the rows represent different welfare statistics of interest. Observed is the equilibrium allocation with all measured barriers. Frictionless is the counterfactual in which all barriers (taxes, political risk, and geopolitical distance) have been removed. Columns (3)–(5) illustrate two additional counterfactuals from which only the corresponding distortion is in place. $(\frac{k_i}{\ell_i})$ is the capital stock per employee, while $(\frac{y_i}{\ell_i})$ is output (GDP) per employee. Actual world PPP$ GDP (including countries not in the model) in 2017 was $121 trillion.

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ital allocation efficiency. When both distortions from taxes and political risk are removed (but distortions from geopolitical distance are maintained), GDP is 5.4% lower than in the frictionless scenario. This confirms our previous suggestion that distortions from taxes and political risk can (and do) interact with distortions from geopolitical distance. When distortions from geopolitical distance and taxes are removed (but distortions from political risk are maintained), the world GDP loss is 0.4%, which is still sizable, but not nearly as large as the GDP loss induced by cultural, linguistic, and geographic barriers. When only the tax wedge remains, the world GDP loss is 2.8%, indicating that differences in capital taxes represent an important source of misallocation for global capital.

VII.B. Capital and Income Inequality

While the overall effect of these three distortions on allocative efficiency and world GDP appears substantial, their effect on cross-country inequality is even more striking. We can gain a sense of this country heterogeneity by looking at how much these distortions change the distribution of capital and output per employee. When capital misallocation resulting from barriers to international investment are removed, we observe a significant decrease in steady-state dispersion of both capital and output per employee. When moving from the frictionless equilibrium to the observed (distorted) equilibrium, the standard deviation of (log) capital per employee increases by 50.9%, and the standard deviation of log output per employee increases by 22.0%.

When distortions from geopolitical distance alone are maintained, dispersion in log capital per employee is 38.3% higher than in the frictionless benchmark. The dispersion of log output per employee is 15.7% higher. Finally, we find that by only maintaining investment taxes (political risk), dispersion in log capital per employee is 13.0% (7.4%) higher compared with the frictionless benchmark, while dispersion in log output per employee is 5.9% (3.9%) higher. In other words, distortions from investment taxes, political risk, and geopolitical distance all significantly contribute to creating long-term cross-country inequality.

Figure VI illustrates the effect of removing distortions from taxes, political risk, and geopolitical distance on cross-country inequality. It shows how the (fitted) cross-country distribution of capital per employee and output per employee changes in re-

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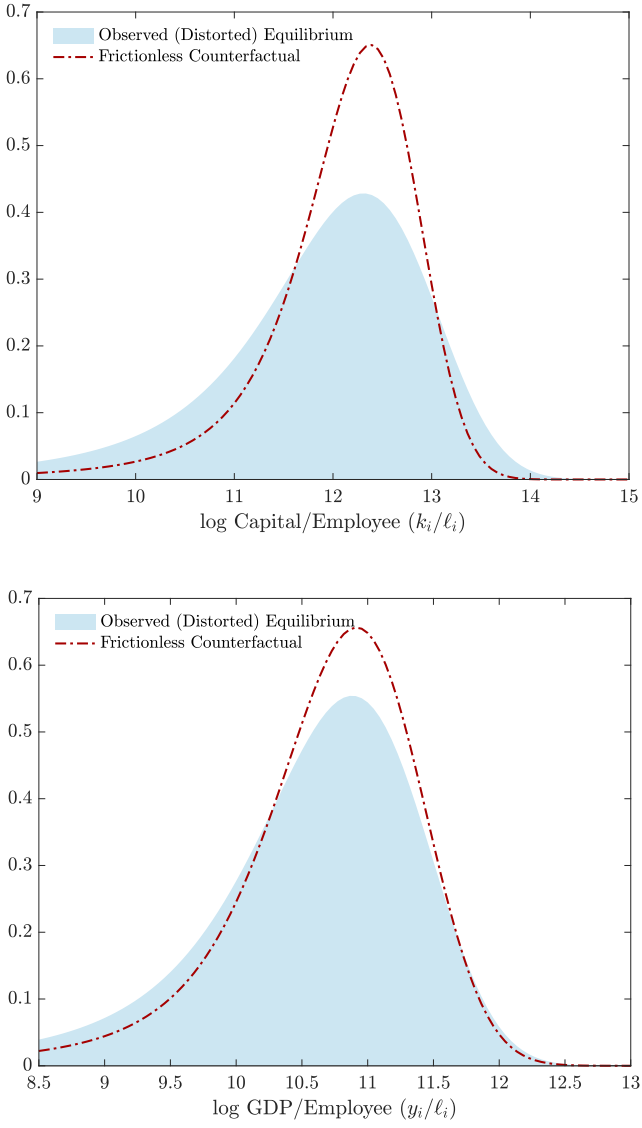

FIGURE VI

Distribution of Capital and Output per Employee

This figure fits the probability density function of a stable distribution (a four-parameter family of distributions with flexible skewness and fat tails) to country-level capital stock per employee (upper panel) and GDP per employee (bottom panel). In each panel, the lighter area is the distribution in the observed, distorted equilibrium. The dotted black line is the distribution in a counterfactual scenario in which all wedges have been removed.

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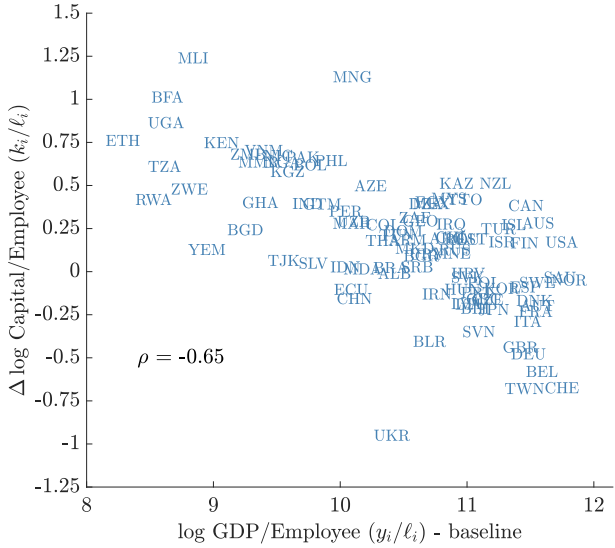

FIGURE VII

Capital Reallocation Gains by Country

This figure displays the baseline level of log output per capita (horizontal axis), against the log change in capital stock per employee (left axis) following the removal of all measured barriers to global capital allocation. Each observation is a country, and the data are from 2017.

sponse to the removal of the barriers. For both variables, we observe a significant reduction in dispersion, but also in skewness (the left tail becomes thinner). We notice a general rightward shift, reflecting an increase in capital and income per employee for the median country.

What explains this reduction in inequality? When capital distortions are removed, capital tends to be reallocated to countries that have higher rates of returns to capital under the distorted equilibrium. As discussed previously, these tend to be countries with lower capital stock per employee and lower output per employee. Figure VII illustrates this effect: it is a scatter plot of the baseline level of GDP per employee (horizontal axis) against the log change in capital per employee from moving to a frictionless world (vertical axis). As can be seen from the graph, there are significant winners and losers—albeit on average most countries experience an increase in capital and output per capita. The strong negative correlation between country-level gains and the initial

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level of output per employee implies that the removal of barriers leads to a substantial reduction in cross-country inequality. Some of the poorest countries see capital per employee increase by an order of magnitude, and income per employee double.

In sum, using counterfactual analysis, we find that the persistent misallocation of capital across countries—induced by investment taxes, political risk, and geopolitical distances—imposes quantitatively important output losses for the majority of countries, and in general for world GDP, and can potentially account for a significant share of the observed cross-country dispersion in capital/employee.[28]

VII.C. Net Positions

Finally, we consider a comparison of net foreign asset positions under the observed equilibrium and the frictionless equilibrium ($\Delta_{ij} = \tau_{ij} = 1$ for all i, j). We define a country i's net foreign asset position as the difference between the invested wealth (s_i) and the capital stock (k_i), and present it as a fraction of GDP.

A notable feature of our model is that it generates persistent (steady-state) global imbalances. Figure VIII displays scatterplots of the resulting net foreign assets against log GDP per employee (middle panel). Under the observed equilibrium, there are large deviations in net investment positions (IIP); yet these net asset positions only correlate weakly with the level of development. This is consistent with Lucas's observation that capital fails to flow from rich to poor countries. It is also consistent with the actual data from the World Bank/PWT. When frictions are removed (right panel), the relationship becomes much stronger in magnitude, as the absolute value of the correlation between net foreign assets and the level of development doubles. In the frictionless equilibrium, capital indeed flows from rich to poor countries. The presence of distortions from taxes, political risk, and geopolitical distance can thus help explain the lack of a strong correlation, in the data, between a country's net asset positions and its level of development.

28. These findings are consistent with those in Portes and Rey (2005), although ours is the first publication to document them in the context of a structural model.

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FIGURE VIII

Net Asset Positions and Development

The figure above plots the model-implied international investment position (IIP) as a share of GDP (y_i), against the log of GDP per employee. The left panel plots IIP/GDP as measured by Lane and Milesi-Ferretti (2018)'s database. The middle panel shows the model-implied IIP/GDP in the observed distorted equilibrium, and the right panel plots IIP/GDP in the frictionless counterfactual, in which all barriers are removed.

VIII. EXTENSIONS AND ROBUSTNESS CHECKS

In Online Appendix G we present several extensions and robustness checks to our baseline model and results. Here we provide a brief summary of these additional analyses, which validate our main findings. We also explore the implications of relaxing a number of assumptions.

In Online Appendix G.1 we examine the short-term effects of removing barriers to international capital allocation under conditions that do not assume a steady-state equilibrium. The results of this exercise suggest that although the full benefits of removing barriers to international investment may take time to materialize, there are already significant gains to be realized in the short term. In particular, the effects on inequality are similar in the short term and in the long term, suggesting that the distributional effect of removing barriers would be relatively immediate and persistent.

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We extend our basic framework in three different directions, to account for international trade frictions, capital controls, and currency hedging costs. First, in Online Appendix G.2 we add frictions to trade of goods across countries and repeat the counterfactual analysis for this extended model. Adding international trade frictions lowers the estimates of aggregate output gains but does not substantially alter the main conclusions about the effect of barriers to global capital allocation.[29] In Online Appendix G.3 we add capital controls by explicitly defining and measuring a capital controls wedge. When we repeat the counterfactual analysis for this extended model, we obtain very similar percentage GDP losses compared to those from the baseline model. The effects of distortions from geopolitical distances are also similar to those in the baseline exercise. In Online Appendix G.4 we parsimoniously incorporate currencies in our framework by modeling currency hedging costs directly. Again, the world GDP losses and inequality effects from repeating our counterfactual exercise in this extended model are basically identical to those from the baseline model, and so are the marginal effects of geopolitical distances, taxes, and political risk.

In Online Appendix G.5 we provide evidence that our coefficient estimates for cultural distance, linguistic distance, and geographical distance are stable over time.

In Online Appendix G.6 we consider alternative measures of international investment by modifying our empirical analysis in two ways. First, while in our main estimation we broke down foreign assets into debt and equity components, we now consider another conventional breakdown of capital positions: foreign direct investment (FDI) and foreign portfolio investment (FPI). We find that cultural, geographic, and linguistic distances exert statistically significant and economically meaningful negative effects on FDI and FPI, with or without additional controls. Second, we replicate the regressions of Table II using non-restated (residency-based) data on foreign total investment, foreign debt investment, and foreign equity investment. The

29. Another interesting extension, which we do not pursue in this article, would be to include asymmetric preferences over consumption goods, with a more complex asset demand system. Such asymmetries, in principle, could generate a further source of home bias, in addition to that analyzed in our baseline framework.

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standardized magnitudes of these estimates are very close to those of Table II.

In Online Appendix G.7 we use alternative estimates to calibrate the semi-elasticity of foreign investment with respect to cultural, linguistic, and geographical distance. The counterfactual exercises based on these alternative estimates are similar in magnitude to those obtained in the baseline using OLS estimates. Finally, in Online Appendix G.8 we briefly discuss how including government bonds in the IMF data may affect our results.

IX. Conclusions

We offered a new theory of international capital allocation: a multi-country dynamic spatial general equilibrium model that features frictions that distort individually rational portfolio allocations.

We showed that a parsimonious implementation of the model—based on just a few explanatory variables—reproduces several key features of international asset markets: (i) it explains a significant share of the observed variation in country portfolios; (ii) it produces large, realistic cross-sectional variation in rates of return across countries, which correlates negatively with the degree of home bias and the level of economic development, translating into persistent capital misallocation; (iii) it predicts, out of sample and with high accuracy, the overall level and the cross section of home bias across countries (where "out-of-sample" means that we do not use any direct information about domestic capital investment); and (iv) it produces steady-state capital account imbalances that do not correlate negatively with the level of development, implying that capital fails to flow from rich to poor countries as much as it should (Lucas 1990).

To quantify the influence of these factors on the international allocation of capital and their real impact, we performed a number of counterfactual exercises. We studied how world GDP and the cross-country distribution of capital and output per worker would change if the effects of barriers to foreign investment were neutralized. This quantitative exercise suggests that these barriers have a sizable effect on the distribution of capital across countries, with implications for efficiency. World GDP is about 7.0% lower than it would be if the effect of these barriers to global capital allocation could be neutralized.

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This misallocation also has significant effects on world inequality. The cross-country standard deviation of capital per employee is 50.9% higher, while the dispersion of output per employee is 22.0% higher than under the frictionless counterfactual. The hypothetical removal (or offsetting) of geopolitical distances, taxes, and political risk would bring substantial economic gains and reductions in cross-country inequality: it would lead capital to reallocate from richer countries, where the rate of return on capital is lower, to poorer countries, where the rate of return is higher. Thus, removing these barriers to the international movement of capital benefits countries that are otherwise "peripheral"—that is, countries that are less accessible to most investors because of these barriers. In sum, these barriers generate and perpetuate an advantage, in terms of capital market access, for "central" countries, and a disadvantage for economies that are geographically and culturally farther from where most investors are located.

While diversification and hedging have generally been viewed as crucial for understanding these patterns, our analysis suggests that international investment frictions also play an important role. How to address these inefficiencies in an effective and coordinated way remains an open area of inquiry.

Our results also have implications for global tax policy coordination. In the presence of geopolitical frictions, the simple harmonization of capital tax rates across countries fails to improve capital allocation efficiency and could even worsen it. From a normative perspective, we find that a social planner aiming to maximize world GDP should impose a lower tax rate on capital in countries that are remote with respect to investors to counterbalance the effect of the geopolitical frictions.

Our study contributes to the literature on open-economy financial macroeconomics by making theoretical and empirical progress in modeling international asset markets in a structural multi-country setting. It also connects to the macroeconomics literature on resource misallocation, by studying the real effects of international asset market frictions. In 1990, Robert Lucas asked, "Why doesn't capital flow from rich to poor countries?" This article sheds new light on this question. Taxes and geopolitical barriers are important determinants of cross-country portfolios and have a major effect on capital allocation efficiency and income distribution.

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Supplementary Material

An Online Appendix for this article can be found at *The Quarterly Journal of Economics* online.

Data Availability

The data underlying this article are available in the Harvard Dataverse, https://doi.org/10.7910/DVN/SBRVPQ (Pellegrino, Spolaore, and Wacziarg 2025).

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